Filtri

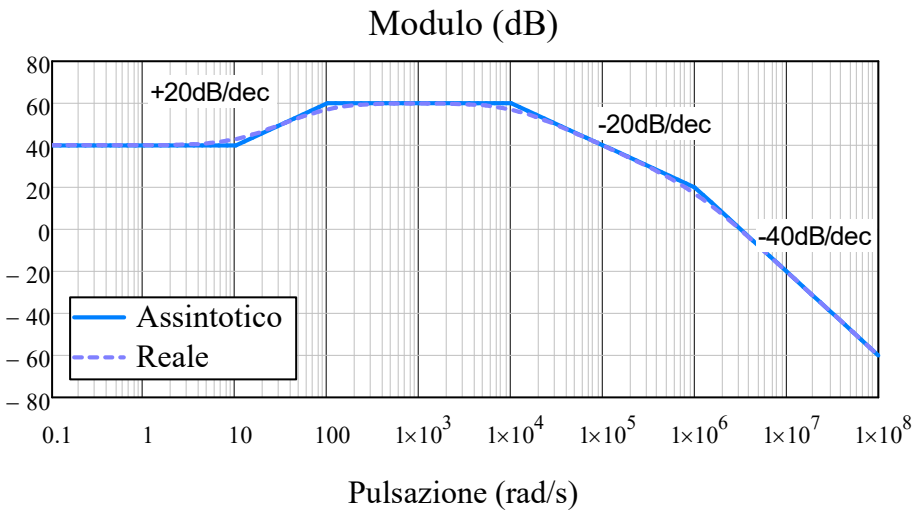
Esercizio 1A

DATI: $A = 100$, $\omega_Z = 10 \cdot rad \cdot s^{-1}$, $\omega_{P_1} = 100 rad \cdot s^{-1}$,

$\omega_{P_2} = 10^4 \cdot rad \cdot s^{-1}$, $\omega_{P_3} = 10^6 \cdot rad \cdot s^{-1}$

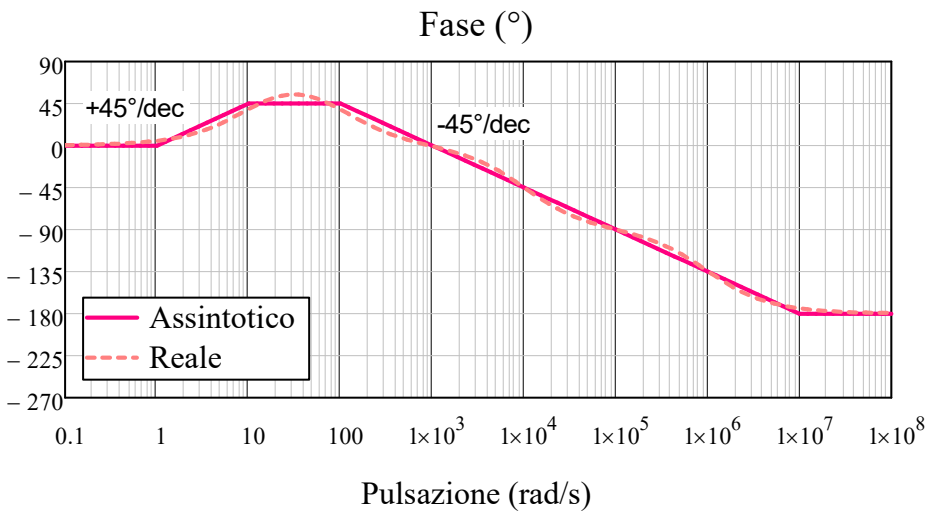
$$W(\omega) = A \cdot \frac{1 + j \cdot \frac{\omega}{\omega_Z}}{\left(1 + j \cdot \frac{\omega}{\omega_{P_1}}\right) \cdot \left(1 + j \cdot \frac{\omega}{\omega_{P_2}}\right) \cdot \left(1 + j \cdot \frac{\omega}{\omega_{P_3}}\right)}$$

Nessun polo e nessuno zero nell'origine. Guadagno positivo. Il diagramma di bode del modulo parte da 40dB.
Per il modulo la pendenza aumenta di 20dB/dec ad ogni zero e diminuisce di 20dB/dec ad ogni polo



Il diagramma di bode della fase parte da 0 poichè non ci sono poli nè zeri nell'origine e A>0

- in corrispondenza dello zero lo sfasamento inizia una decade prima e finisce una decade dopo. Complessivamente aumenta di 90° (45° per ogni decade) tra $\omega_Z/10$ fino a $10\omega_Z$.
- in corrispondenza di ciascun polo lo sfasamento inizia una decade prima e finisce una decade dopo. Complessivamente diminuisce di 90° (45° per ciascuna decade) tra $\omega_P/10$ fino a $10\omega_P$.



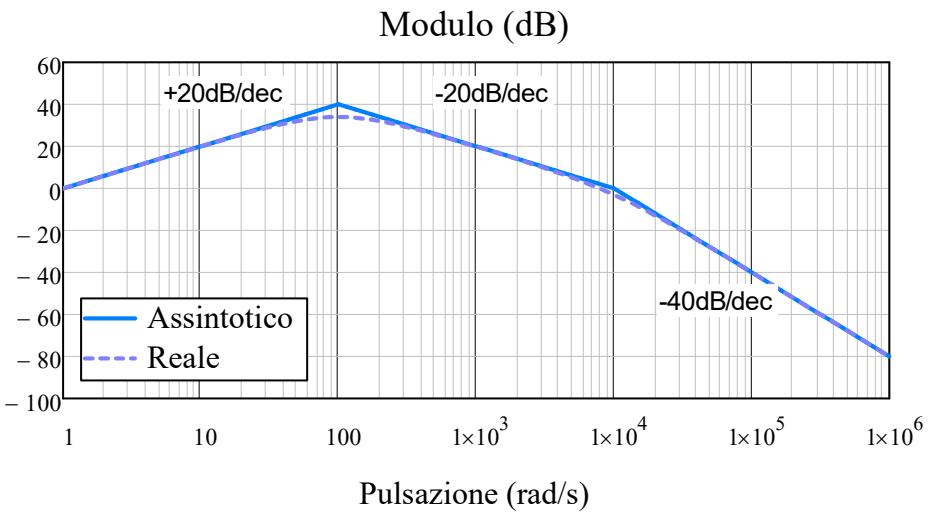
Sfasamento iniziale (0°)							
ω_{Z1}	+45	+45					
ω_{P1}		-45	-45				
ω_{P2}				-45	-45		
ω_{P3}						-45	-45
pendenza totale:	+45	0	-45	-45	-45	-45	-45

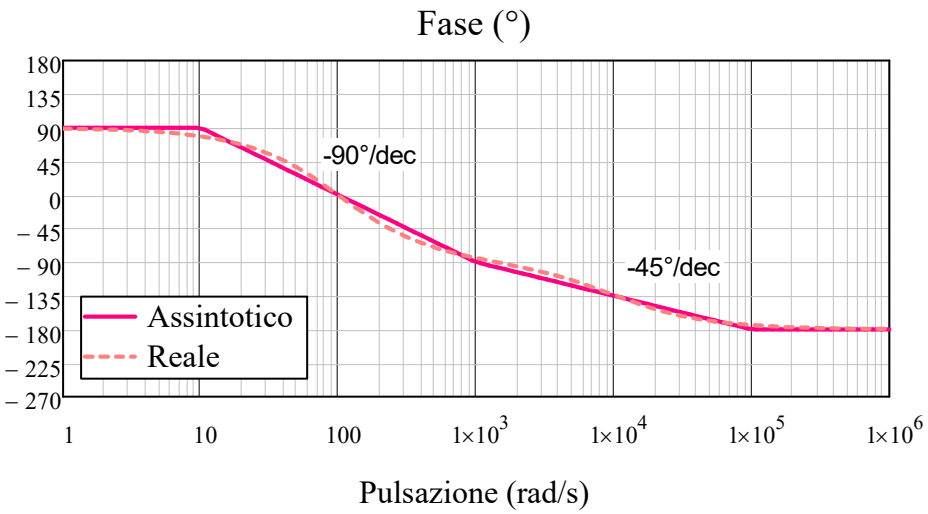
Esercizio 1B

DATI: $A = 1,\ \omega_O = 1s^{-1}\ \omega_{P_1} = 100 rad \cdot s^{-1},\ \omega_{P_2} = 10^4 \cdot rad \cdot s^{-1}$

$$W(\omega) = A \cdot \frac{j \cdot \frac{\omega}{\omega_O}}{\left(1 + j \cdot \frac{\omega}{\omega_{P_1}}\right)^2 \cdot \left(1 + j \cdot \frac{\omega}{\omega_{P_2}}\right)}$$

Il diagramma di bode del modulo parte con una pendenza di +20dB/dec poichè c'è uno zero nell'origine.
Il diagramma di bode della fase parte da +90° poichè il guadagno è positivo e c'è un polo nell'origine.





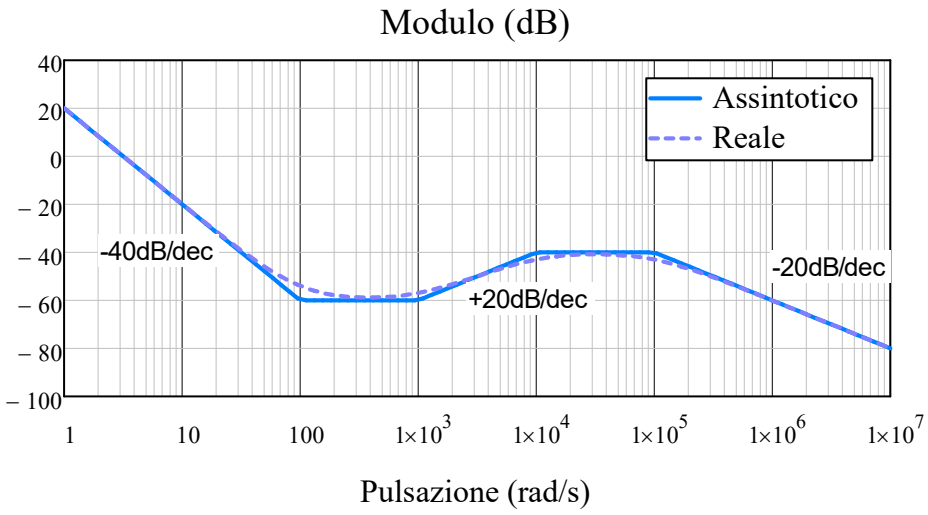
Sfasamento (°/dec)				
ω_{P1} (polo doppio)	-90°/dec	-90°/dec		
ω_{P2}			-45°/dec	-45°/dec
pendenza totale:	-90°/dec	-90°/dec	-45°/dec	-45°/dec

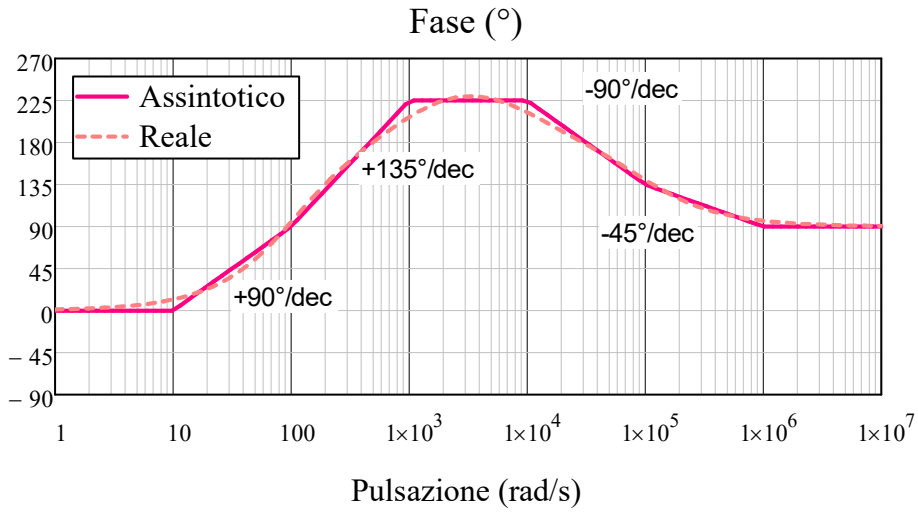
Esercizio 1C

DATI: $A = -10$, $\omega_0 = 1 rad \cdot s^{-1}$, $\omega_{Z_1} = 100 rad \cdot s^{-1}$, $\omega_{Z_2} = 10^3 rad \cdot s^{-1}$, $\omega_{P_1} = 10^4 rad \cdot s^{-1}$, $\omega_{P_2} = 10^5 \cdot rad \cdot s^{-1}$

$$W(\omega) = A \cdot \frac{\left(1 + \frac{j \cdot \omega}{\omega_{Z_1}}\right)^2 \cdot \left(1 + \frac{j \cdot \omega}{\omega_{Z_2}}\right)}{\left(\frac{j \cdot \omega}{\omega_0}\right)^2 \cdot \left(1 + \frac{j \cdot \omega}{\omega_{P_1}}\right) \cdot \left(1 + \frac{j \cdot \omega}{\omega_{P_2}}\right)}$$

Il diagramma di bode del modulo parte con una pendenza di -40dB per decade poichè c'è un polo doppio nell'origine

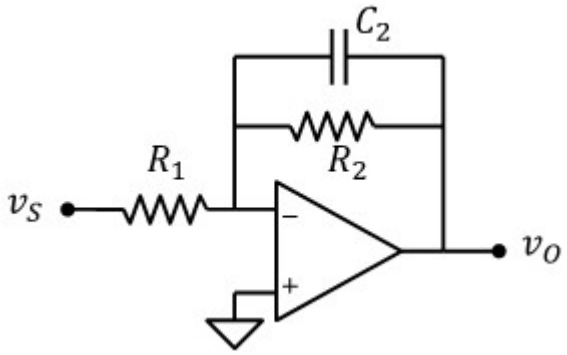




ω_{Z1} (zero doppio)	+90°/dec	+90°/dec			
ω_{Z2}		+45°/dec	+45°/dec		
ω_{P1}			-45°/dec	-45°/dec	
ω_{P2}				-45°/dec	-45°/dec
pendenza totale:	+90°/dec	+135°/dec	0	-90°/dec	-45°/dec

Esercizio 2

DATI: $R_1 = 10k\Omega$, $R_2 = 5k\Omega$, $C_2 = 200nF$



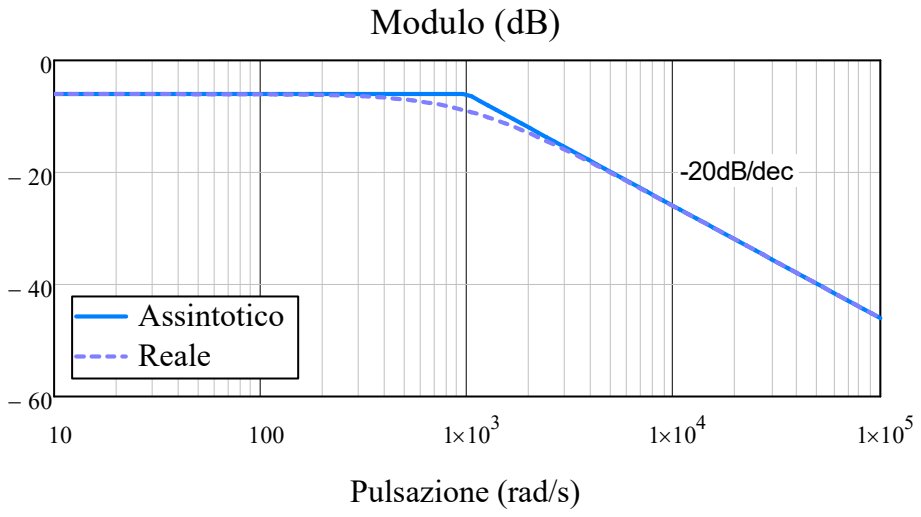
Funzione di trasferimento

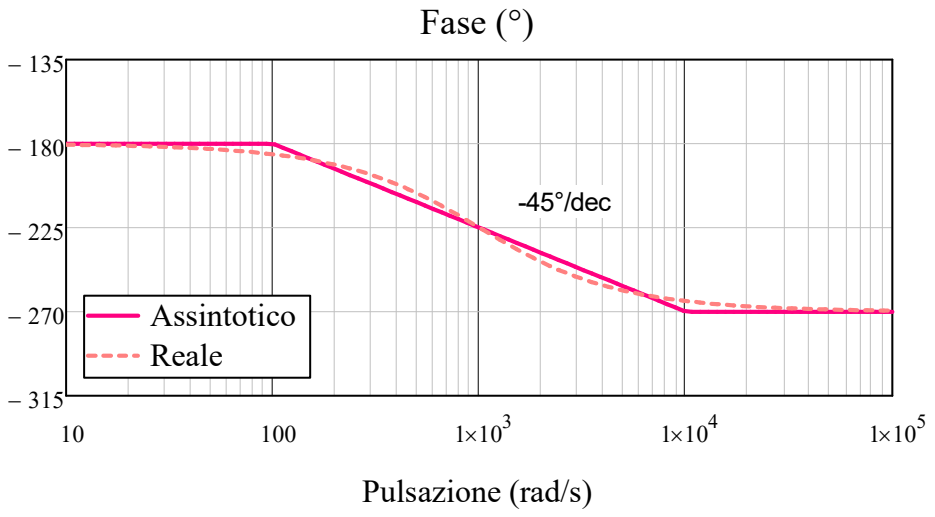
$$Z_2 = \frac{R_2}{1 + j\omega \cdot R_2 \cdot C_2}$$

$$v_O = -v_S \cdot \frac{Z_2}{R_1} = \frac{-R_2}{R_1} \cdot \frac{1}{\left(1 + j\omega \cdot R_2 \cdot C_2\right)} \cdot v_S$$

Definiamo: $\boxed{A = \frac{-R_2}{R_1} = -0.5}$ $\boxed{\omega_P = \frac{1}{C_2 \cdot R_2} = 1 \times 10^3 \cdot s^{-1}}$ $\boxed{W(\omega) = A \cdot \frac{1}{\left(1 + \frac{j\omega}{\omega_P}\right)}}$

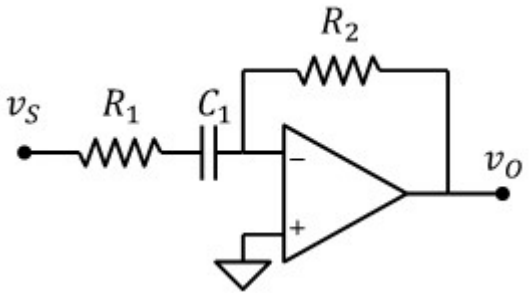
Diagramma di Bode asintotico del modulo e della fase





Esercizio 3

DATI: $R_1 = 10k\Omega$, $R_2 = 50k\Omega$, $C_1 = 10nF$



Funzione di trasferimento

$$Z_1 = \frac{1 + j\omega \cdot R_1 \cdot C_1}{j\omega \cdot C_1}$$

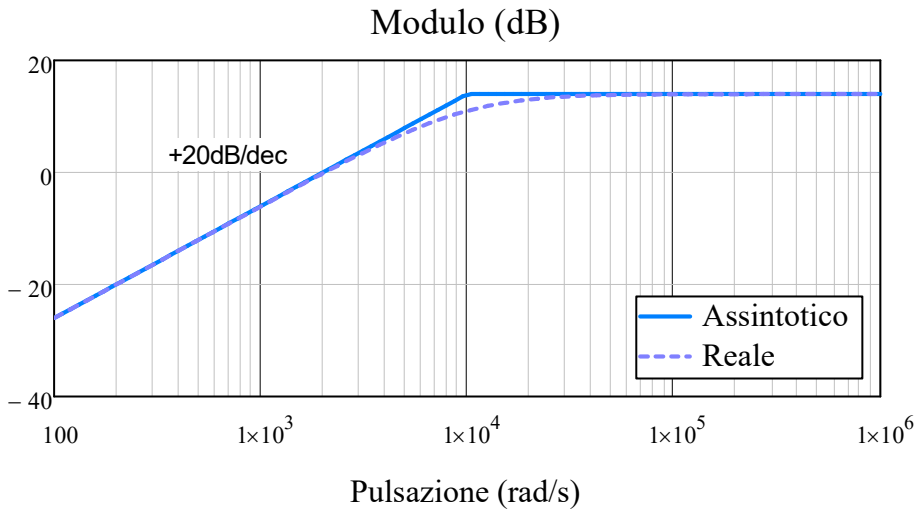
$$v_O = -v_S \cdot \frac{R_2}{Z_1} = \frac{-R_2}{R_1} \cdot \frac{j\omega \cdot C_1 \cdot R_1}{\left(1 + j\omega \cdot R_1 \cdot C_1\right)} \cdot v_S$$

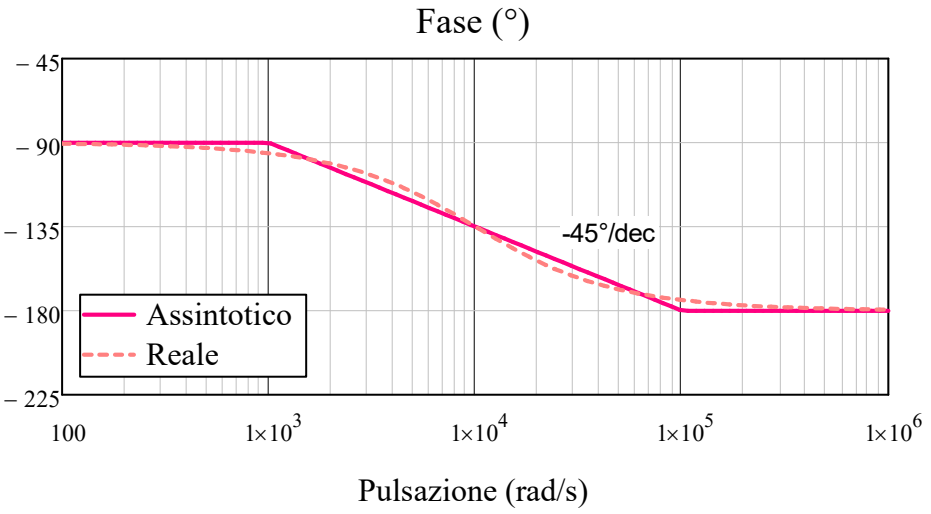
$$\boxed{A = \frac{-R_2}{R_1} = -5}$$

$$\boxed{\omega_P = \frac{1}{C_1 \cdot R_1} = 1 \times 10^4 \cdot s^{-1}}$$

$$\boxed{W(\omega) = A \cdot \frac{\frac{j\omega}{\omega_P}}{\left(1 + \frac{j\omega}{\omega_P}\right)}}$$

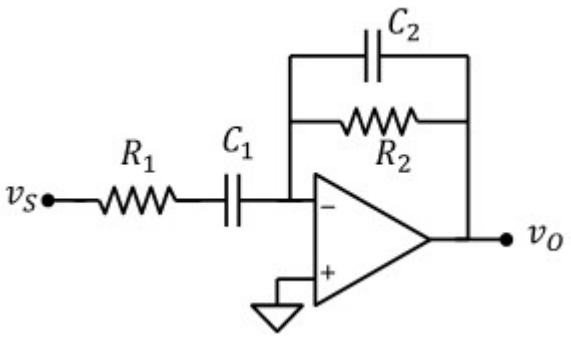
Diagramma di Bode asintotico del modulo e della fase





Esercizio 4

DATI: $R_1 = 10k\Omega$, $R_2 = 5k\Omega$, $C_1 = 10nF$, $C_2 = 2nF$



Funzione di trasferimento

$$Z_1 = \frac{1 + j\omega \cdot R_1 \cdot C_1}{j\omega \cdot C_1} \qquad Z_2 = \frac{R_2}{1 + j\omega \cdot R_2 \cdot C_2}$$

$$v_O = -v_S \cdot \frac{Z_2}{Z_1} = \frac{-R_2}{R_1} \cdot \frac{j\omega \cdot C_1 \cdot R_1}{(1 + j\omega \cdot R_1 \cdot C_1) \cdot (1 + j\omega \cdot R_2 \cdot C_2)} \cdot v_S$$

Definiamo:

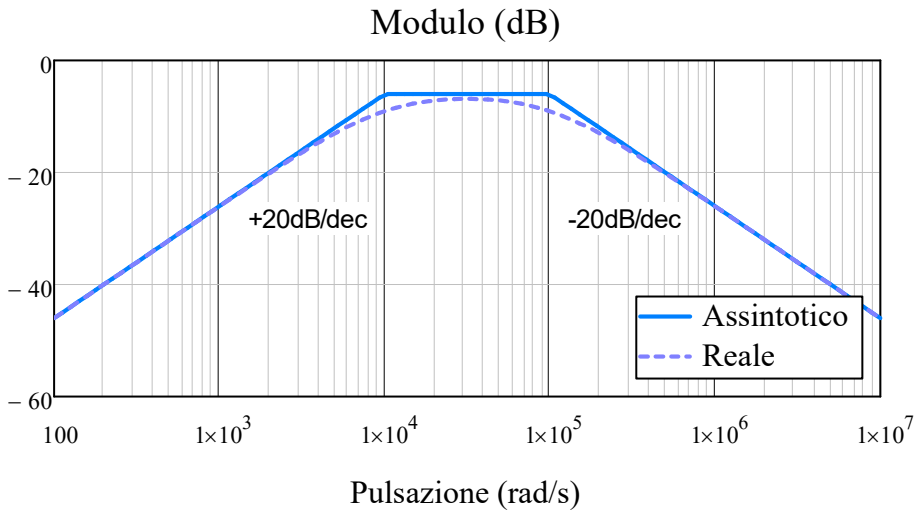
$$A = \frac{-R_2}{R_1} = -0.5$$

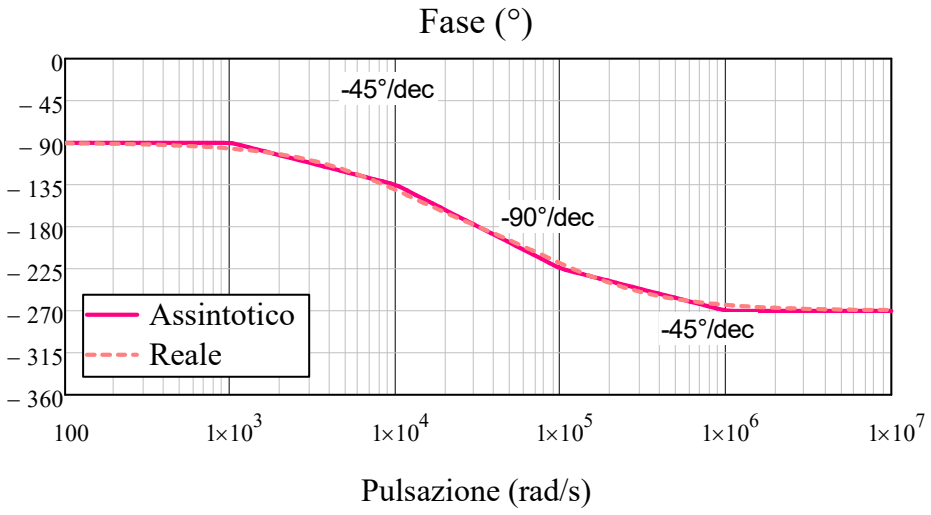
$$\omega_{P_1} = \frac{1}{C_1 \cdot R_1} = 1 \times 10^4 \cdot s^{-1}$$

$$\omega_{P_2} = \frac{1}{C_2 \cdot R_2} = 1 \times 10^5 \cdot s^{-1}$$

$$W(\omega) = A \cdot \frac{\frac{j\omega}{\omega_{P_1}}}{\left(1 + \frac{j\omega}{\omega_{P_1}}\right) \cdot \left(1 + \frac{j\omega}{\omega_{P_2}}\right)}$$

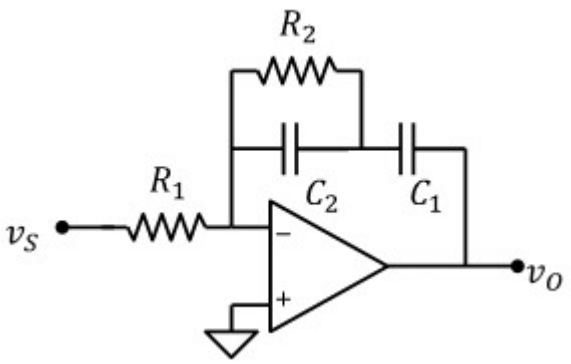
Diagramma di Bode asintotico del modulo e della fase





Esercizio 5

DATI: $R_1 = 1k\Omega$, $R_2 = 10k\Omega$, $C_1 = 900nF$, $C_2 = 100nF$



Funzione di trasferimento

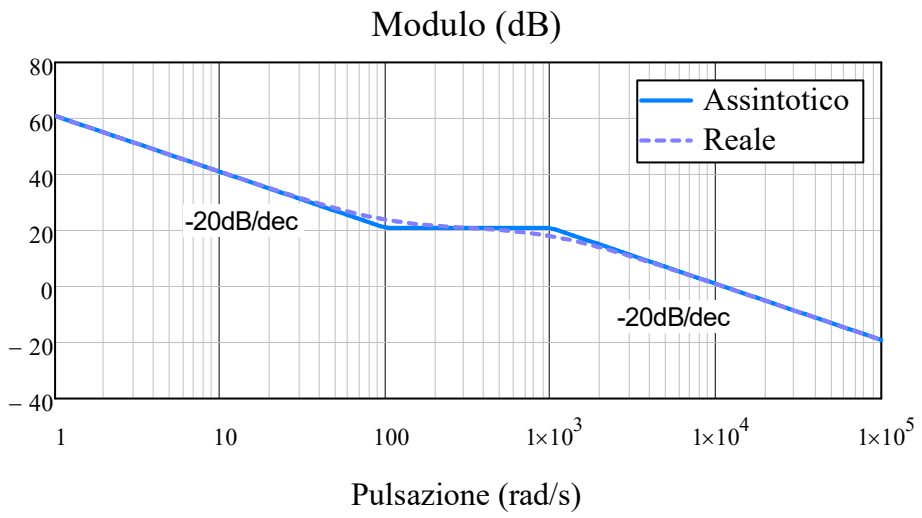
$$Z_2 = \frac{1}{j\omega \cdot C_1} + \frac{R_2}{1 + j\omega \cdot R_2 \cdot C_2} = \frac{1 + j\omega \cdot R_2 \cdot (C_1 + C_2)}{j\omega \cdot C_1 \cdot (1 + j\omega \cdot R_2 \cdot C_2)}$$

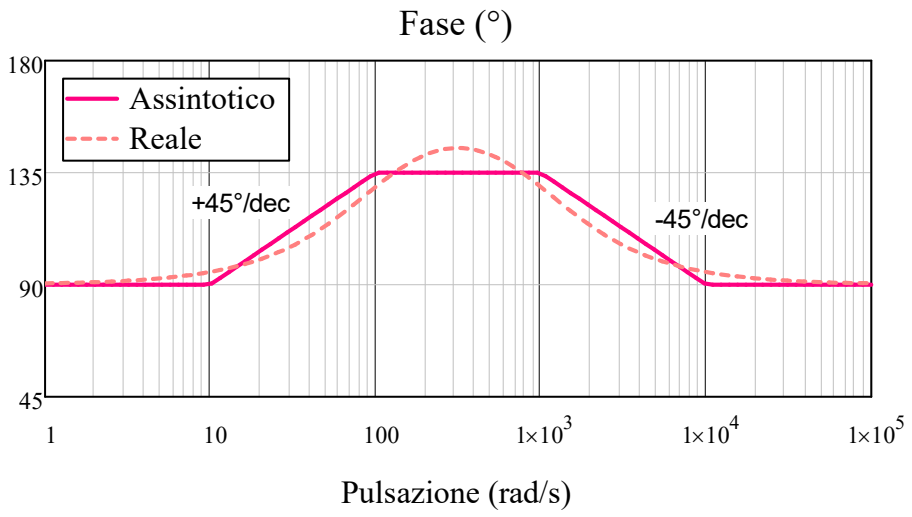
$$v_O = -\frac{Z_2}{R_1} = -\frac{1 + j\omega \cdot R_2 \cdot (C_1 + C_2)}{j\omega \cdot C_1 \cdot R_1 \cdot (1 + j\omega \cdot R_2 \cdot C_2)}$$

Definiamo: $\boxed{\omega_Z = \left[R_2 \cdot (C_1 + C_2)\right]^{-1} = 100 \cdot s^{-1}}$ $\boxed{\omega_P = (R_2 \cdot C_2)^{-1} = 1 \times 10^3 \cdot s^{-1}}$ $\boxed{\omega_O = (C_1 \cdot R_1)^{-1} = 1.11 \times 10^3 \cdot s^{-1}}$

$$\boxed{W(\omega) = \frac{-\left(1 + \frac{j\omega}{\omega_Z}\right)}{\frac{j\omega}{\omega_O} \cdot \left(1 + \frac{j\omega}{\omega_P}\right)}}$$

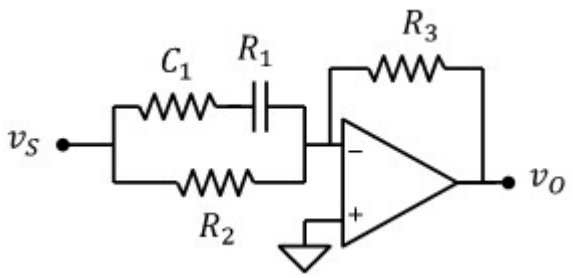
Diagramma di Bode asintotico del modulo e della fase





Esercizio 6

DATI: $R_1 = 1k\Omega$, $R_2 = 9k\Omega$, $R_3 = 45k\Omega$, $C_1 = 1\mu F$



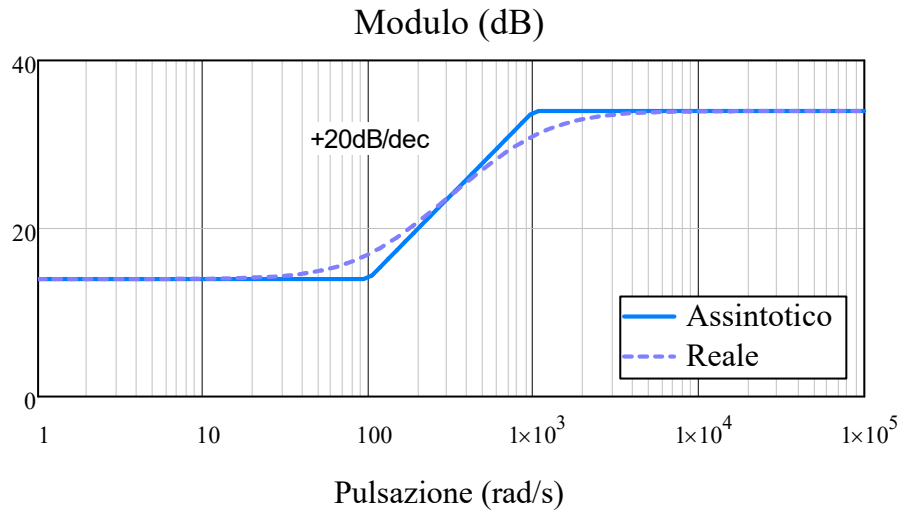
Funzione di trasferimento

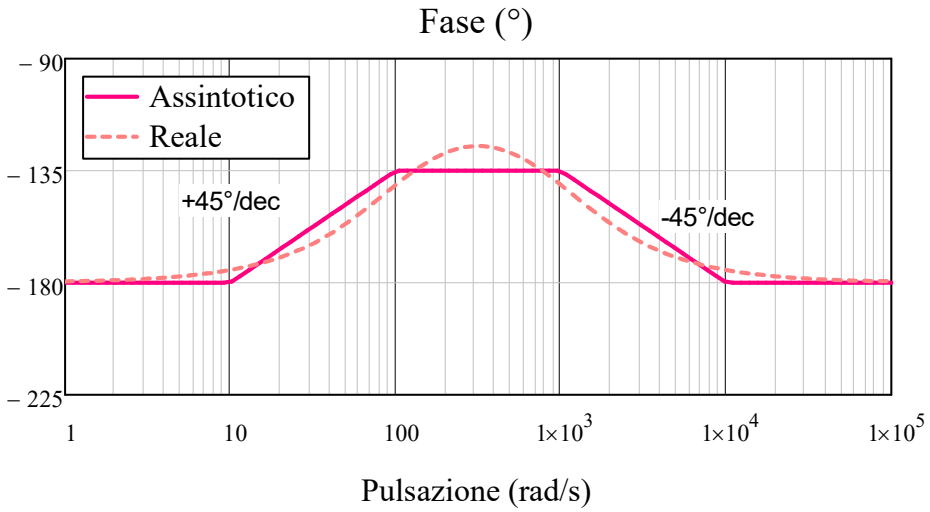
$$v_O = \frac{-R_3}{Z} \qquad Z = \frac{R_2 \cdot Z_1}{R_2 + Z_1} \qquad Z_1 = \frac{1 + j\omega \cdot R_1 \cdot C_1}{j\omega \cdot C_1}$$

$$v_O = \frac{-R_3 \cdot (R_2 + Z_1)}{R_2 \cdot Z_1} = \frac{-R_3}{R_2} \cdot \left(1 + \frac{j\omega \cdot C_1 \cdot R_2}{1 + j\omega \cdot R_1 \cdot C_1}\right) = \frac{-R_3}{R_2} \cdot \left[\frac{1 + j\omega \cdot (R_1 + R_2) \cdot C_1}{1 + j\omega \cdot R_1 \cdot C_1}\right]$$

$$A = \frac{-R_3}{R_2} = -5 \qquad \omega_Z = \frac{1}{(R_1 + R_2) \cdot C_1} = 100 \cdot s^{-1} \qquad \omega_P = \frac{1}{R_1 \cdot C_1} = 1 \times 10^3 \cdot s^{-1} \qquad W(\omega) = A \cdot \frac{1 + \frac{j\omega}{\omega_Z}}{1 + \frac{j\omega}{\omega_P}}$$

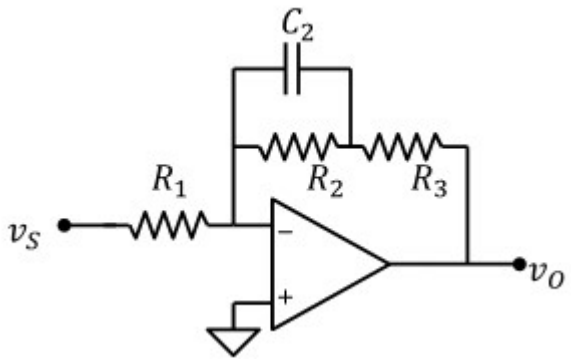
Diagramma di Bode asintotico del modulo e della fase





Esercizio 7

DATI: $R_1 = 2k\Omega$, $R_2 = 198k\Omega$, $R_3 = 2k\Omega$, $C_2 = 50.5nF$



Funzione di trasferimento

$$Z_2 = \frac{R_2}{1 + j\omega \cdot R_2 \cdot C_2}$$

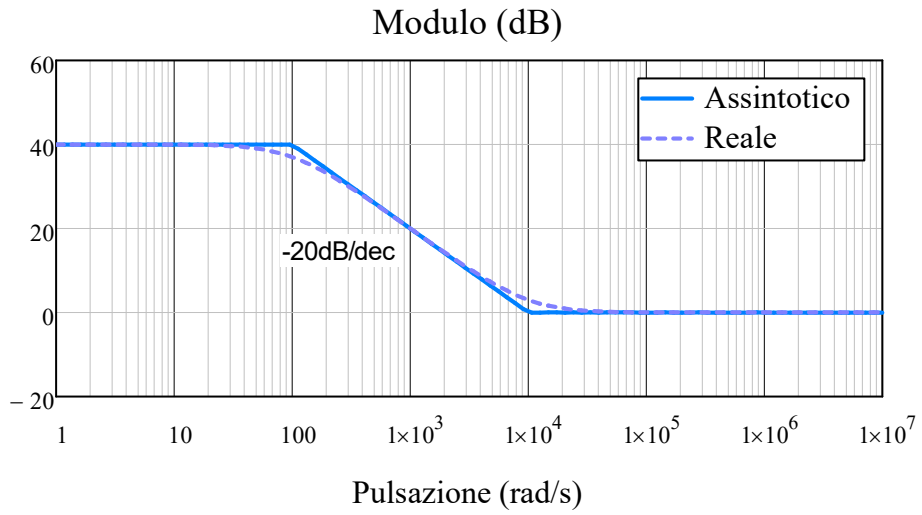
$$v_O = -\frac{R_3 + Z_2}{R_1} = -\left(\frac{R_2 + R_3}{R_1}\right) \cdot \frac{1 + j\omega \cdot \frac{R_3 \cdot R_2}{R_2 + R_3} \cdot C_2}{1 + j\omega \cdot R_2 \cdot C_2}$$

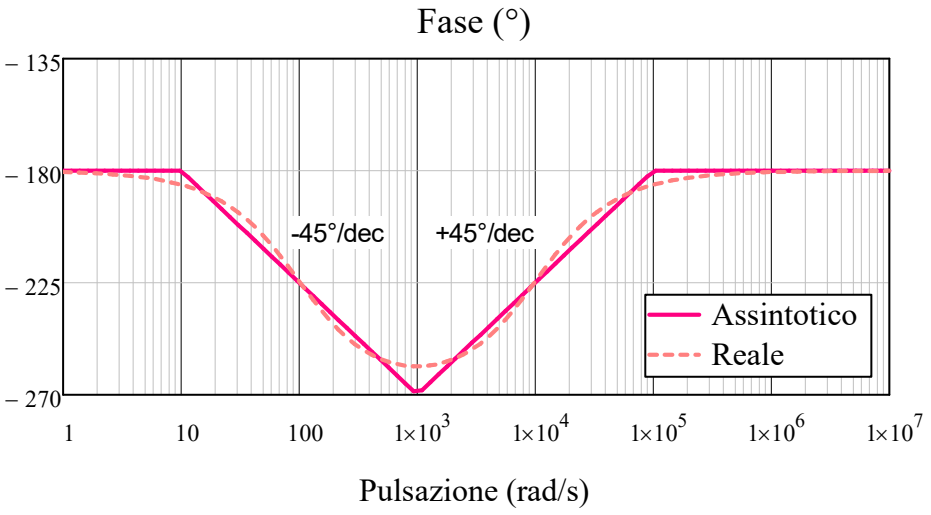
Definiamo:

$$R_P = \frac{R_3 \cdot R_2}{R_2 + R_3} = 1.98 \cdot k\Omega \qquad \boxed{\omega_Z = \frac{1}{C_2 \cdot R_P} = 1 \times 10^4 \cdot s^{-1}} \qquad \boxed{\omega_P = \frac{1}{C_2 \cdot R_2} = 100 \cdot s^{-1}} \qquad \boxed{A = -\left(\frac{R_2 + R_3}{R_1}\right) = -100}$$

$$\boxed{W(\omega) = A \cdot \frac{1 + \frac{j\omega}{\omega_Z}}{1 + \frac{j\omega}{\omega_P}}}$$

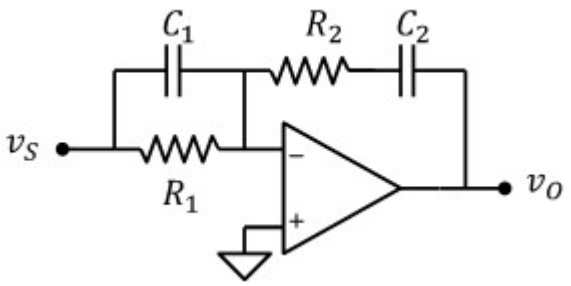
Diagramma di Bode asintotico del modulo e della fase





Esercizio 8

DATI: $R_1 = 10k\Omega$, $R_2 = 5k\Omega$, $C_1 = 1nF$, $C_2 = 20nF$



Funzione di trasferimento

$$Z_1 = \frac{R_1}{1 + j\omega \cdot R_1 \cdot C_1} \qquad Z_2 = \frac{1 + j\omega \cdot R_2 \cdot C_2}{j\omega \cdot C_2}$$

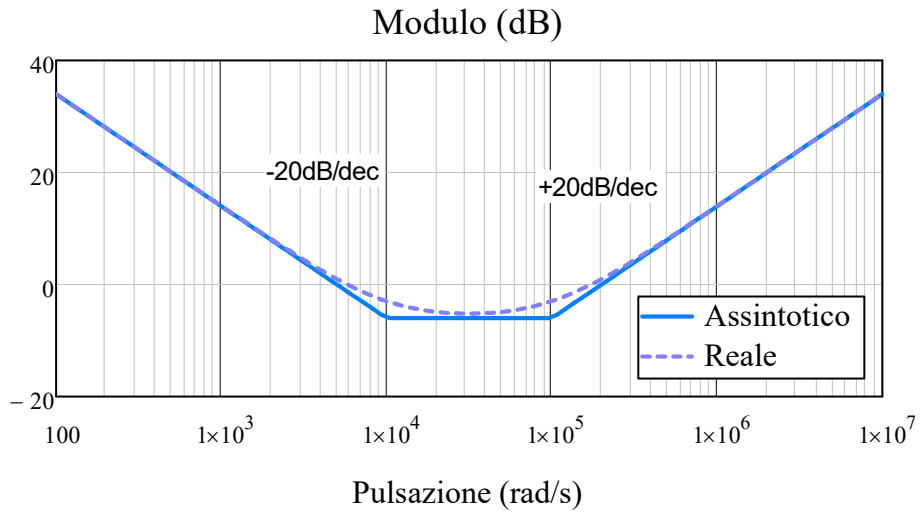
$$v_O = -v_S \cdot \frac{Z_2}{Z_1} = \frac{-R_2}{R_1} \cdot \frac{(1 + j\omega \cdot R_2 \cdot C_2) \cdot (1 + j\omega \cdot R_1 \cdot C_1)}{j\omega \cdot C_2 \cdot R_2} \cdot v_S$$

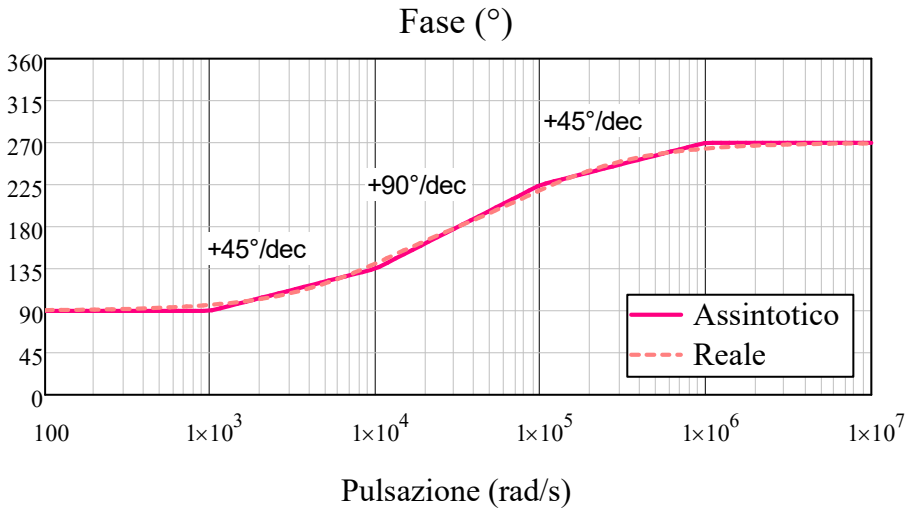
Definiamo:

$$A = \frac{-R_2}{R_1} = -0.5 \qquad \omega_{Z_1} = \frac{1}{C_1 \cdot R_1} = 1 \times 10^5 \cdot s^{-1} \qquad \omega_{Z_2} = \frac{1}{C_2 \cdot R_2} = 1 \times 10^4 \cdot s^{-1}$$

$$W(\omega) = A \cdot \frac{\left(1 + \frac{j\omega}{\omega_{Z_1}}\right) \cdot \left(1 + \frac{j\omega}{\omega_{Z_2}}\right)}{\frac{j\omega}{\omega_{Z_2}}}$$

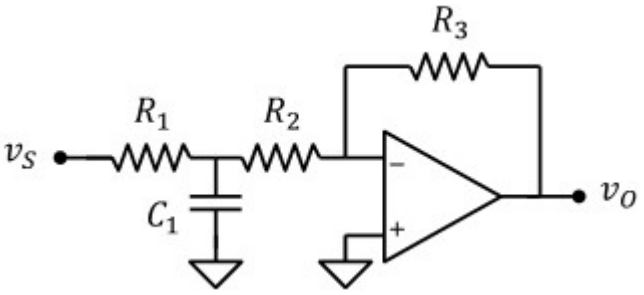
Diagramma di Bode asintotico del modulo e della fase





Esercizio 9

DATI: $R_1 = 1k\Omega$, $R_2 = 1k\Omega$, $R_3 = 200k\Omega$, $C_1 = 20nF$



Funzione di trasferimento

Rete equivalente secondo thevening all'ingresso invertente dell'AO

Tensione a vuoto: $$v_{EQ} = v_S \cdot \frac{1}{1 + j\omega \cdot R_1 \cdot C_1}$$

Resistenza equivalente (con v_S=0):

$$Z_{EQ} = R_2 + \frac{R_1}{1 + j\omega \cdot R_1 \cdot C_1} = \frac{R_2 + R_1 + j\omega \cdot R_1 \cdot R_2 \cdot C_1}{\left(1 + j\omega \cdot R_1 \cdot C_1\right)} = \left(R_2 + R_1\right) \cdot \frac{1 + j\omega \cdot \frac{R_1 \cdot R_2}{R_2 + R_1} \cdot C_1}{\left(1 + j\omega \cdot R_1 \cdot C_1\right)}$$

Configurazione invertente:

$$v_O = \frac{-R_3}{Z_{EQ}} \cdot v_{EQ} = -v_S \cdot \frac{1}{1 + j\omega \cdot R_1 \cdot C_1} \cdot \frac{R_3 \cdot \left(1 + j\omega \cdot R_1 \cdot C_1\right)}{\left(R_2 + R_1\right) \cdot \left(1 + j\omega \cdot \frac{R_1 \cdot R_2}{R_2 + R_1} \cdot C_1\right)} = -v_S \cdot \frac{R_3}{\left(R_2 + R_1\right)} \cdot \frac{1}{\left(1 + j\omega \cdot \frac{R_1 \cdot R_2}{R_2 + R_1} \cdot C_1\right)}$$

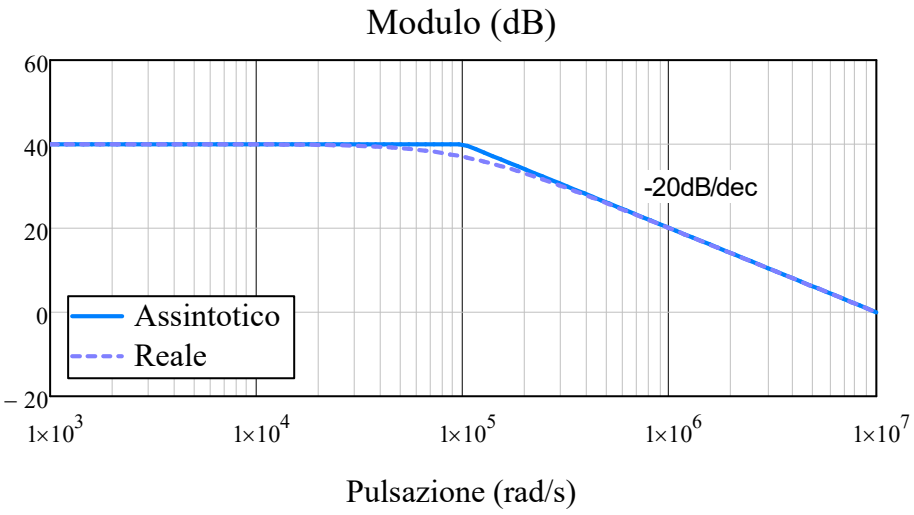
Poniamo: $$\boxed{A = \frac{-R_3}{R_1 + R_2} = -100}$$

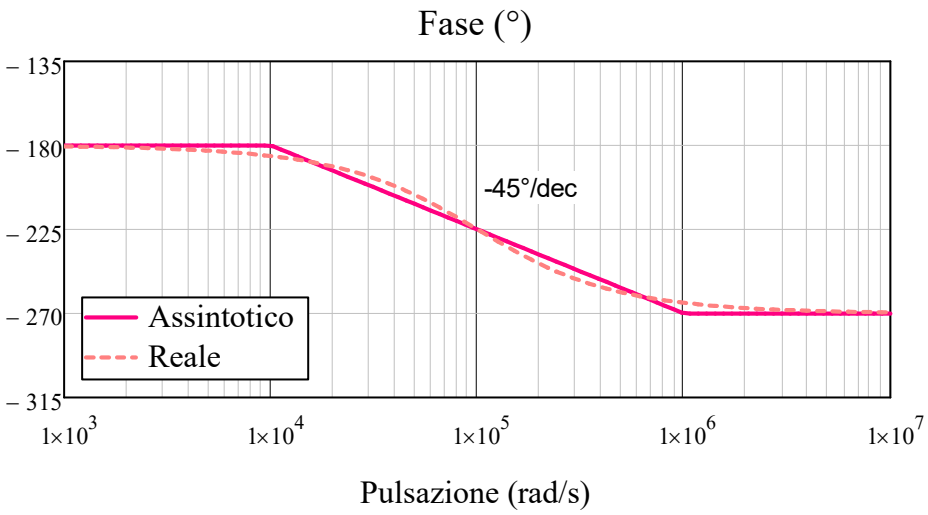
$$R_P = \frac{R_1 \cdot R_2}{R_2 + R_1} = 500\,\Omega \qquad \boxed{\omega_P = \left(R_P \cdot C_1\right)^{-1} = 1 \times 10^5 \cdot s^{-1}}$$

Otteniamo la funzione di trasferimento:

$$\boxed{W(\omega) = \frac{A}{1 + \frac{j\omega}{\omega_P}}}$$

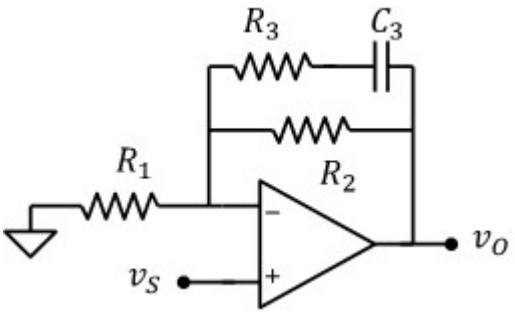
Diagramma di Bode asintotico del modulo e della fase





Esercizio 10

DATI: $R_1 = 1k\Omega$, $R_2 = 99k\Omega$, $R_3 = 1k\Omega$, $C_3 = 100nF$



1) Guadagno in condizioni stazionarie

$$\boxed{A_v = 1 + \frac{R_2}{R_1} = 100}$$

2) Guadagno ad alta frequenza

$$R_P = \frac{R_2 \cdot R_3}{R_2 + R_3} \qquad \boxed{A_v = 1 + \frac{R_P}{R_1} = 1.99}$$

2) Ricavare l'espressione della funzione di trasferimento

$$Z_2 = \frac{R_2 \cdot \left(R_3 + \frac{1}{j\omega \cdot C_3}\right)}{R_2 + \left(R_3 + \frac{1}{j\omega \cdot C_3}\right)} = \frac{R_2 \cdot (j\omega \cdot C_3 \cdot R_3 + 1)}{j\omega \cdot C_3 \cdot (R_2 + R_3) + 1}$$

$$W(\omega) = 1 + \frac{Z_2}{R_1} = 1 + \frac{R_2 \cdot (j\omega \cdot C_3 \cdot R_3 + 1)}{R_1 \cdot [j\omega \cdot C_3 \cdot (R_2 + R_3) + 1]} = \frac{R_1 \cdot [j\omega \cdot C_3 \cdot (R_2 + R_3) + 1] + R_2 \cdot (j\omega \cdot C_3 \cdot R_3 + 1)}{R_1 \cdot [j\omega \cdot C_3 \cdot (R_2 + R_3) + 1]}$$

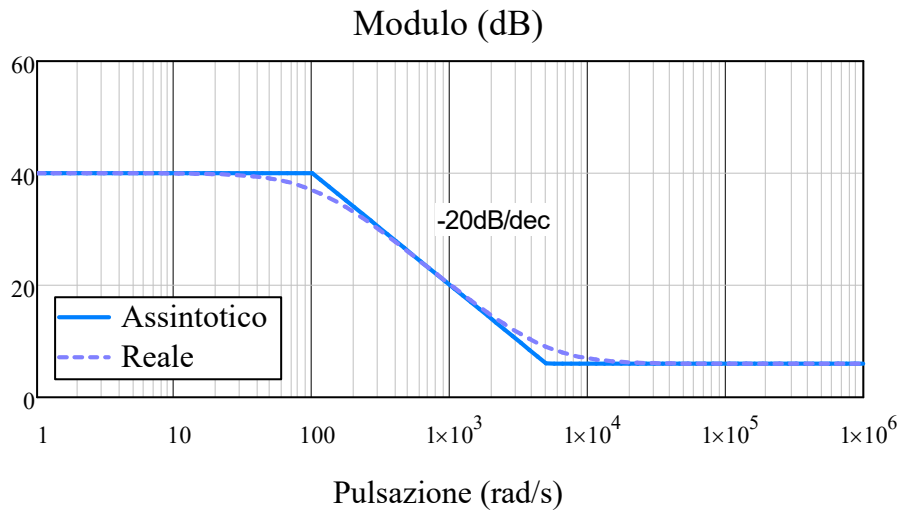
$$W(\omega) = \frac{R_1 + R_2 + j\omega \cdot C_3 \cdot (R_2 \cdot R_1 + R_3 \cdot R_1 + R_2 \cdot R_3)}{R_1 \cdot [1 + j\omega \cdot C_3 \cdot (R_2 + R_3)]} = \left(1 + \frac{R_2}{R_1}\right) \cdot \frac{1 + j\omega \cdot C_3 \cdot \left(\frac{R_2 \cdot R_1}{R_1 + R_2} + R_3\right)}{1 + j\omega \cdot C_3 \cdot (R_2 + R_3)}$$

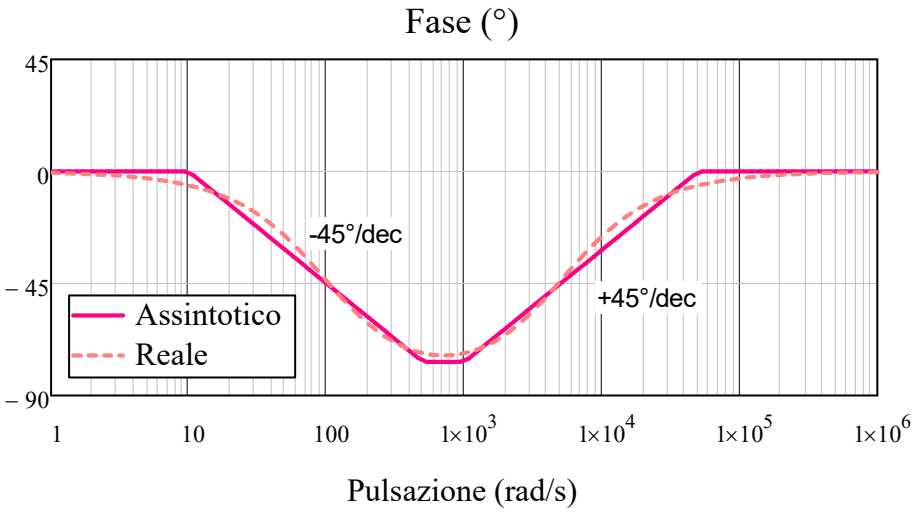
Poniamo: $A = \left(1 + \frac{R_2}{R_1}\right) = 100$ $\qquad \omega_{Z_1} = \frac{1}{C_3 \cdot \left(\frac{R_2 \cdot R_1 + R_3 \cdot R_1 + R_2 \cdot R_3}{R_1 + R_2}\right)} = 5 \times 10^3 \cdot s^{-1}$

$$\omega_{P_1} = \frac{1}{C_3 \cdot (R_2 + R_3)} = 100 \cdot s^{-1} \qquad \frac{R_2 \cdot R_1 + R_3 \cdot R_1 + R_2 \cdot R_3}{R_1 + R_2} = 1.99 \times 10^3\,\Omega$$

$$W(\omega) = A \cdot \frac{1 + \frac{j \cdot \omega}{\omega_{Z_1}}}{1 + \frac{j \cdot \omega}{\omega_{P_1}}}$$

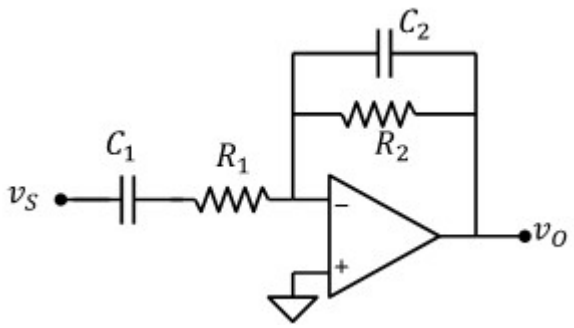
3) Tracciare il diagramma di Bode asintotico del modulo e della fase





Esercizio 11

DATI: $\omega_L = 100 s^{-1}$, $\omega_H = 10^5 \cdot s^{-1}$, $A_v = -10$, $R_{IN} = 10k\Omega$



1) Dimensionamento di resistenze e capacità

Calcolo della funzione di trasferimento:

$$Z_1 = R_1 + \frac{1}{j\omega C_1} = \frac{1 + j\omega \cdot R_1 \cdot C_1}{j\omega \cdot C_1}$$

$$Z_2 = \frac{R_2 \cdot \frac{1}{j\omega \cdot C_2}}{R_2 + \frac{1}{j\omega \cdot C_2}} = \frac{R_2}{1 + j\omega \cdot R_2 \cdot C_2}$$

$$W(\omega) = -\frac{Z_2}{Z_1} = -\frac{j\omega \cdot C_1 \cdot R_2}{(1 + j\omega \cdot R_2 \cdot C_2) \cdot (1 + j\omega \cdot R_1 \cdot C_1)}$$

W(ω) ha uno zero nell'origine e due poli:

I due poli sono: $\omega_{P_1} = \frac{1}{R_1 \cdot C_1}$ $\omega_{P_2} = \frac{1}{R_2 \cdot C_2}$

L'impedenza di ingresso del filtro è $R_{IN} = Z_1$. Ad alta frequenza Z_1 si approssima con R_1, quindi

$$\boxed{R_1 = R_{IN} = 10 \cdot k\Omega}$$

Dobbiamo assumere $\omega_{P1} < \omega_{P2}$. Quindi al centro della banda passante ($\omega_{P1} < \omega < \omega_{P2}$) avremmo:

- $1+j\omega R_1 C_1$ circa uguale a 1
- $1+j\omega R_2 C_1$ circa uguale a 1

il guadagno a centro banda è approssimativamente: $A_v = -\frac{j\omega \cdot C_1 \cdot R_2}{(1) \cdot (j\omega \cdot R_1 \cdot C_1)} = -\frac{R_2}{R_1}$

Bisogna realizzare: $\boxed{R_2 = -A_v \cdot R_1 = 100 \cdot k\Omega}$

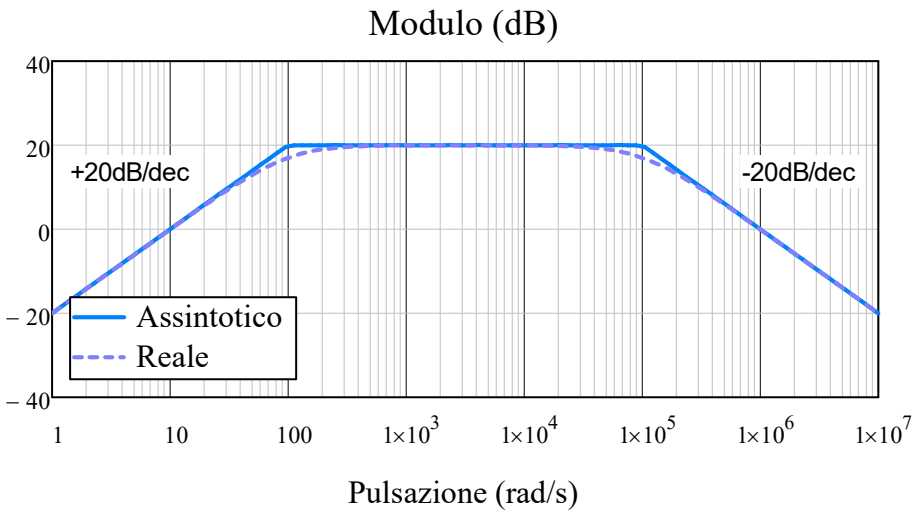
Note le resistenze calcoliamo le capacità:

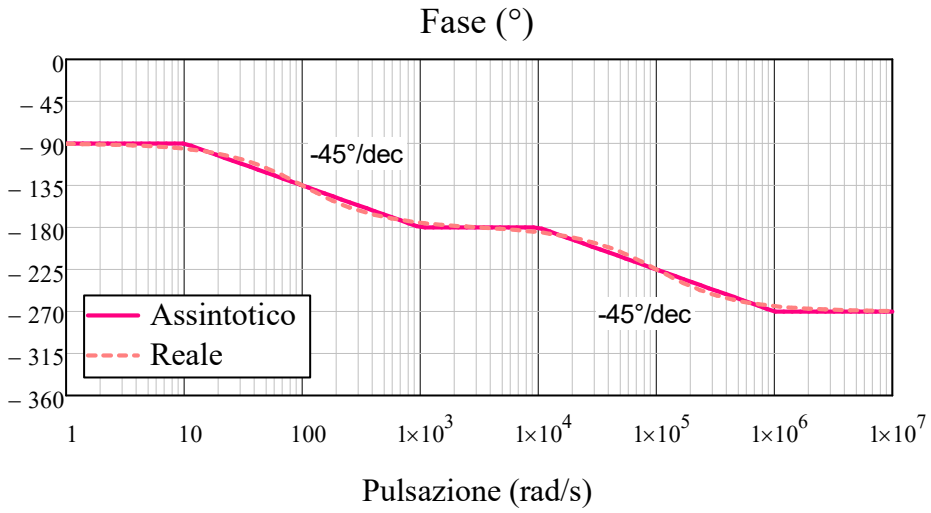
$\omega_{P_1} = \omega_L = 100 \cdot s^{-1}$ $\boxed{C_1 = \frac{1}{R_1 \cdot \omega_{P_1}} = 1 \cdot \mu F}$

$\omega_{P_2} = \omega_H = 1 \times 10^5 \cdot s^{-1}$ $\boxed{C_2 = \frac{1}{R_2 \cdot \omega_{P_2}} = 0.1 \cdot nF}$

$$W(\omega) = \frac{A_v}{\omega_{P_1}} \cdot \frac{j \cdot \omega}{\left(1 + \frac{j \cdot \omega}{\omega_{P_1}}\right) \cdot \left(1 + \frac{j \cdot \omega}{\omega_{P_2}}\right)}$$

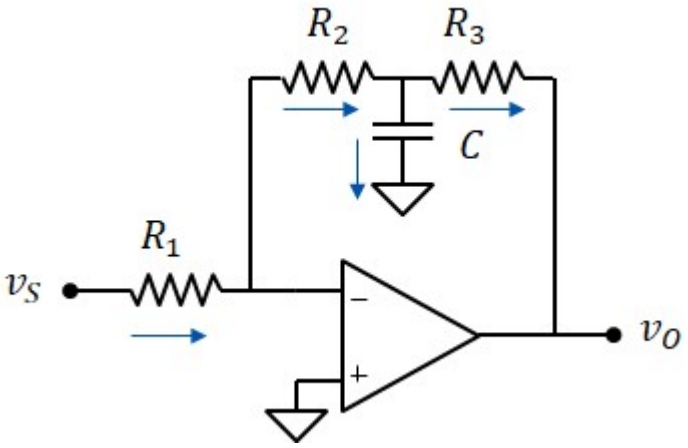
3) Tracciare il diagramma di Bode asintotico del modulo e della fase





Esercizio 12

DATI: $R_1 = 10k\Omega$, $R_2 = 10k\Omega$, $R_3 = 10k\Omega$, $\omega_{Z_1} = 1000s^{-1}$



1) Funzione di trasferimento

Corrente attraverso R_1 e R_2

$$I_{R1} = \frac{v_S}{R_1} \qquad I_{R2} = I_{R1} = \frac{v_S}{R_1}$$

Potenziale ai capi della capacità C:

$$v_C = 0 - V_{R2} = \frac{-R_2}{R_1}\cdot v_S$$

Corrente attraverso C:

$$I_C = \frac{v_C}{Z_C} = j\cdot\omega\cdot C\cdot v_C$$

Corrente attraverso R_3

$$I_{R3} = I_{R2} - I_C = \frac{v_S}{R_1} - j\cdot\omega\cdot C\cdot v_C = \frac{v_S}{R_1} - j\cdot\omega\cdot C\cdot\left(\frac{-R_2}{R_1}\cdot v_S\right) = \left(\frac{1}{R_1} + \frac{j\cdot\omega\cdot C\cdot R_2}{R_1}\right)\cdot v_S$$

Tensione di uscita: $$v_O = v_C - V_{R3} = \frac{-R_2}{R_1}\cdot v_S - R_3\cdot\left(\frac{1}{R_1} + \frac{j\cdot\omega\cdot C\cdot R_2}{R_1}\right)\cdot v_S = -\left(\frac{R_2}{R_1} + \frac{R_3}{R_1} + \frac{j\cdot\omega\cdot C\cdot R_2\cdot R_3}{R_1}\right)\cdot v_S$$

$$W(\omega) = -\frac{R_2 + R_3}{R_1}\cdot\left(1 + j\cdot\omega\cdot C\cdot\frac{R_2\cdot R_3}{R_2 + R_3}\right)$$

Definiamo: $$A = -\frac{R_2 + R_3}{R_1} = -2$$

$$R_P = \frac{R_2\cdot R_3}{R_2 + R_3} = 5\cdot k\Omega$$

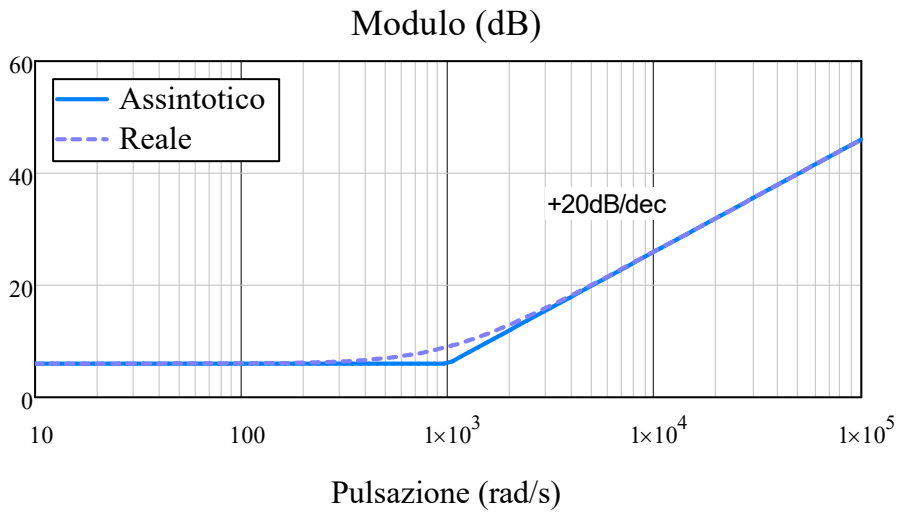
$$\boxed{W(\omega) = A\cdot\left(1 + j\frac{\omega}{\omega_{Z_1}}\right)}$$

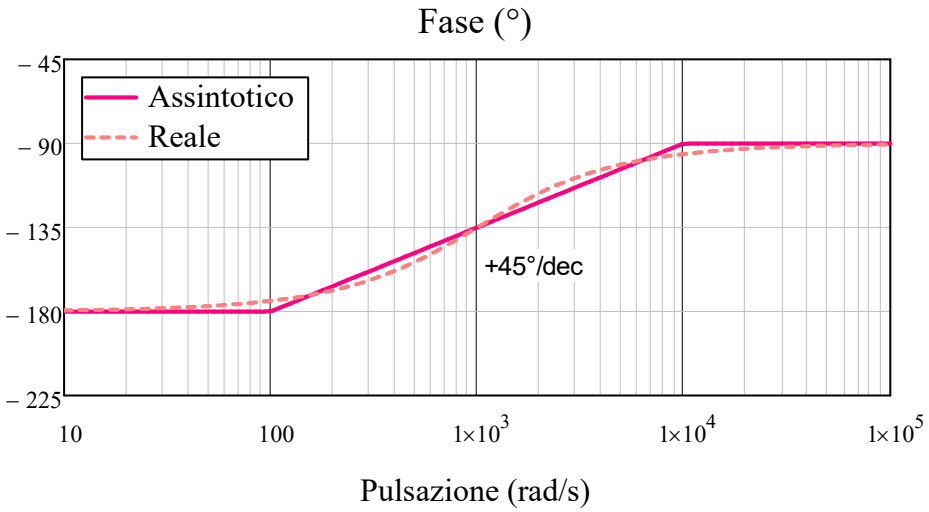
funzione di trasferimento con un solo zero in: $$\omega_{Z_1} = \frac{1}{C\cdot R_P}$$

2) valore di C per avere $\omega_{Z_1} = 1\times 10^3\cdot s^{-1}$

$$\boxed{C = \frac{1}{\omega_{Z_1}\cdot R_P} = 200\cdot nF}$$

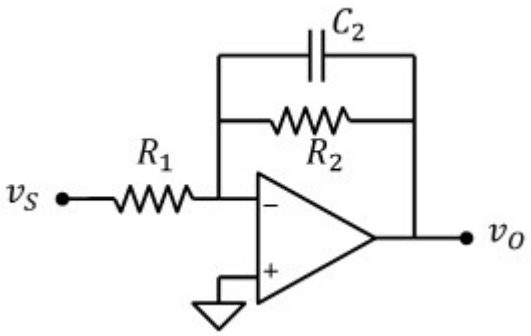
3) Diagramma di Bode asintotico del modulo e della fase





Esercizio 13

DATI: $R_2 = 10k\Omega$



1) valore di R_1 affinchè il modulo del guadagno sia $W_0 = 5$

$$A_v = \frac{-R_2}{R_1}$$ (filtro passa basso)

$$\boxed{R_1 = \frac{R_2}{W_0} = 2 \cdot k\Omega}$$

2) valore di C_2 affinchè la frequenza di taglio sia $\omega_T = 10^3 s^{-1}$

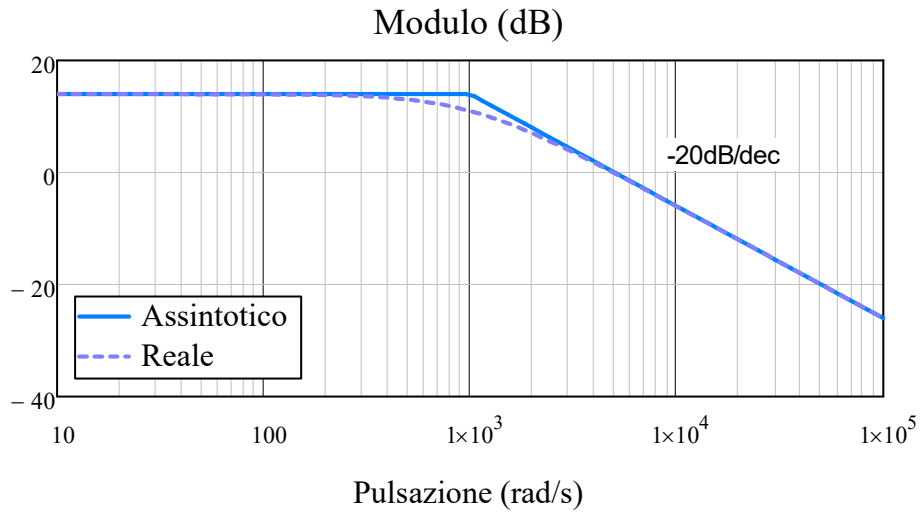
$$\omega_T = (R_2 \cdot C_2)^{-1}$$

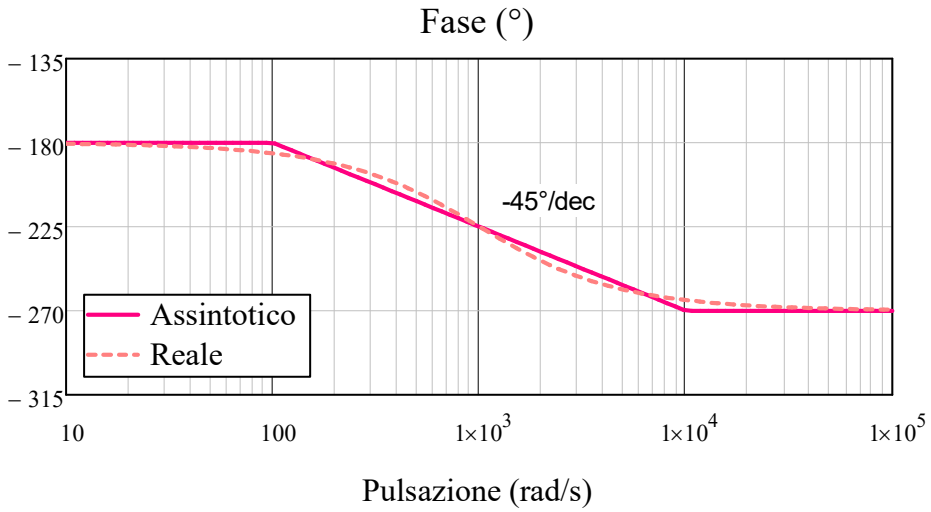
$$\boxed{C_2 = \frac{1}{R_2 \cdot \omega_T} = 100 \cdot nF}$$

3) Diagramma di Bode asintotico del modulo e della fase

Funzione di trasferimento: $Z_2 = \dfrac{R_2}{1 + j\omega \cdot R_2 \cdot C_2}$ $\quad W(\omega) = \dfrac{-Z_2}{R_1} = \dfrac{-R_2}{R_1} \cdot \dfrac{1}{1 + j\omega \cdot R_2 \cdot C_2}$

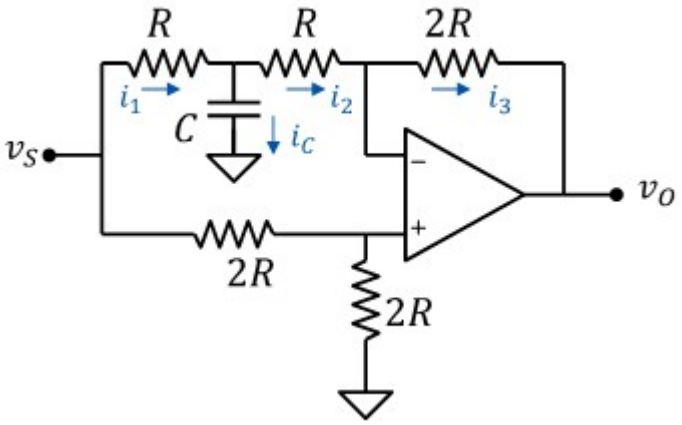
$$A = \frac{-R_2}{R_1} = -5 \qquad \omega_{P_1} = \frac{1}{R_2 \cdot C_2} = 1 \times 10^3 \cdot s^{-1} \qquad W(\omega) = \frac{A}{1 + \frac{j\omega}{\omega_{P_1}}}$$





Esercizio 14

DATI: $R = 2k\Omega$, $C = 100nF$
Segnale di ingresso: $V_S = 5V$, $\varphi_S = 45°$



1) Funzione di trasferimento

$$v_P = \frac{v_S}{2} = v_N$$

Dalle leggi di kirchhoff ricaviamo:

$$i_1 = i_C + i_2 \qquad i_2 = i_3$$

Sia v_C la tensione ai capi del condensatore:

$$i_1 = \frac{v_S - v_C}{R} \qquad i_2 = \frac{v_C - \frac{v_S}{2}}{R} \qquad i_3 = \frac{v_O - \frac{v_S}{2}}{2R} \qquad i_C = v_C \cdot j\omega \cdot C$$

$$\frac{v_S - v_C}{R} = \frac{v_C - \frac{v_S}{2}}{R} + v_C \cdot j\omega \cdot C$$

Ricaviamo v_C:

$$v_C = \frac{3v_S}{2\cdot(2 + j\omega\cdot R\cdot C)}$$

Calcoliamo $i_3 = i_2$

$$i_3 = i_2 = \frac{v_C - \frac{v_S}{2}}{R} = \frac{1 - j\omega\cdot R\cdot C}{2 + j\omega\cdot R\cdot C}\cdot\frac{v_S}{2R}$$

e infine v_O:

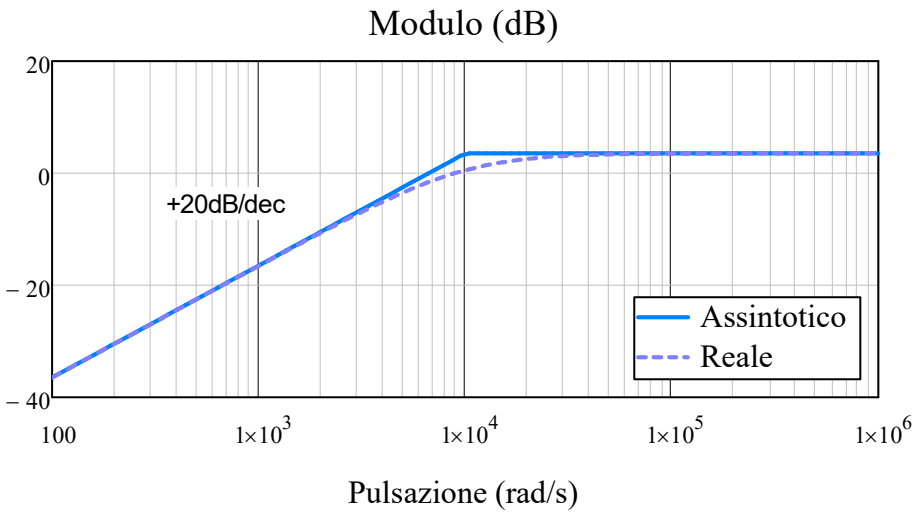
$$v_O = \frac{v_S}{2} - 2R\cdot i_3 = \frac{v_S}{2} - \left(\frac{1 - j\omega\cdot R\cdot C}{2 + j\omega\cdot R\cdot C}\cdot v_S\right) = v_S\cdot\left[\frac{j\omega\cdot 3R\cdot C}{4\cdot\left(1 + j\omega\cdot\frac{R}{2}\cdot C\right)}\right]$$

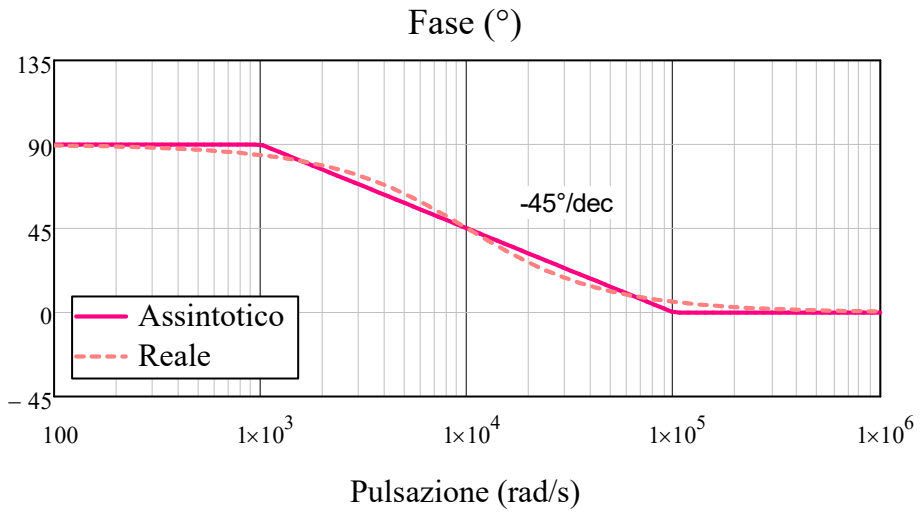
Poniamo: $A = \frac{3}{2} \qquad \omega_{P_1} = \frac{2}{R\cdot C} = 1\times 10^4\cdot s^{-1}$

Otteniamo la seguente funzione di trasferimento:

$$\boxed{W(\omega) = A\cdot\frac{j\frac{\omega}{\omega_{P_1}}}{1 + j\frac{\omega}{\omega_{P_1}}}}$$

2) diagramma di Bode asintotico del modulo e della fase





3a) Ampiezza e fase del segnale di uscita, con segnale di ingresso a pulsazione: $\omega_S = 400s^{-1}$

$$v_S(t) = V_S \cdot \sin(\omega_S \cdot t + \varphi_S)$$

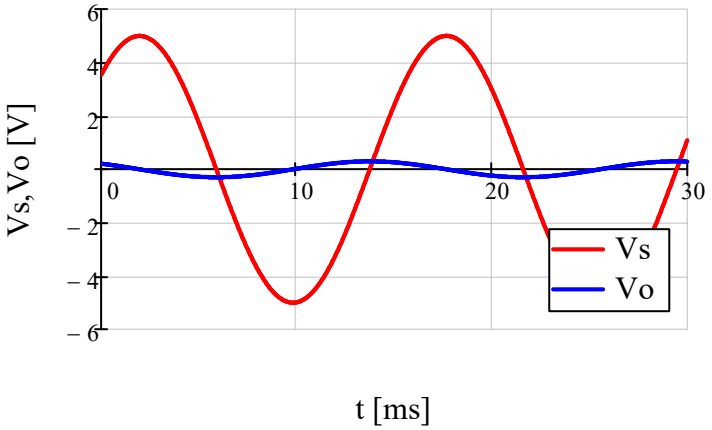
Modulo del guadagno asintotico per $\omega=\omega_S$:

$$A_{S_{dB}} = \left(\left|W(\omega_S)\right|\right)_{dB} = \left(\left|W(\omega_{P_1})\right|\right)_{dB} + 20\cdot\log\left(\frac{\omega_S}{\omega_{P_1}}\right) \qquad A_{S_{dB}} = -24.5dB \qquad A_S = 10^{\frac{A_{S_{dB}}}{20}} = 0.06$$

Fase del guadagno asintotico per $\omega=\omega_S$: $\Delta\varphi_S = 90°$

$$\boxed{V_O = A_S \cdot V_S = 0.3\,V} \qquad \boxed{\varphi_O = \varphi_S + \Delta\varphi_S = 135\cdot°}$$

$$\boxed{v_O(t) = V_O \cdot \sin(\omega_S \cdot t + \varphi_O)}$$



3b) Ampiezza e fase del segnale di uscita, con segnale di ingresso a pulsazione: $\omega_S = 5000 s^{-1}$

Modulo del guadagno asintotico per $\omega=\omega_S$:

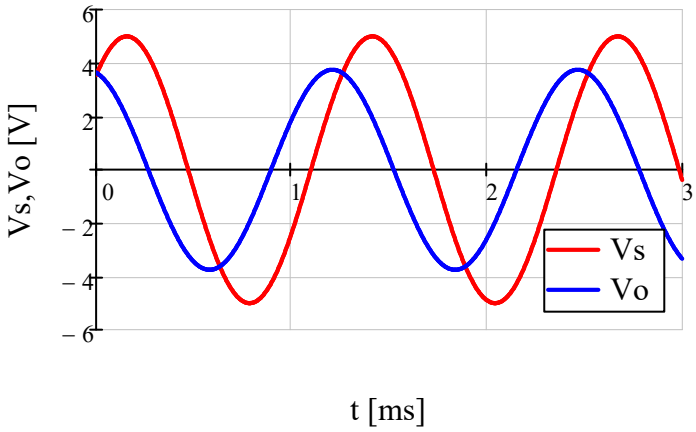
$$A_{S_{dB}} = \left(\left|W(\omega_S)\right|\right)_{dB} = \left(\left|W(\omega_{P_1})\right|\right)_{dB} + 20\cdot\log\left(\frac{\omega_S}{\omega_{P_1}}\right) \qquad A_{S_{dB}} = -2.5dB \qquad A_S = 10^{\frac{A_{S_{dB}}}{20}} = 0.75$$

Fase del guadagno asintotico per $\omega=\omega_S$:

$$\Delta\varphi_S = 90° - 45°\cdot\log\left(\frac{\omega_S}{0.1\,\omega_{P_1}}\right) = 58.5\cdot°$$

$$V_O = A_S\cdot V_S = 3.75\,V \qquad \varphi_O = \varphi_S + \Delta\varphi_S = 103.5\cdot°$$

$$v_O(t) = V_O\cdot\sin(\omega_S\cdot t + \varphi_O)$$



3c) Ampiezza e fase del segnale di uscita, con segnale di ingresso a pulsazione: $\omega_S = 20000 s^{-1}$

Modulo del guadagno asintotico per $\omega=\omega_S$:

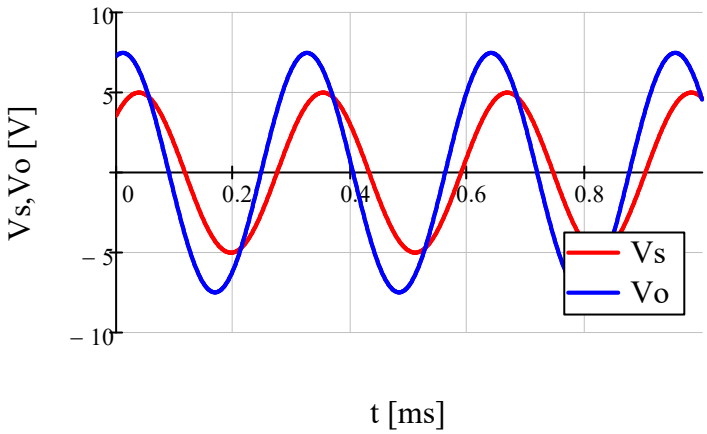
$$A_{S_{dB}} = 3.5dB \qquad A_S = 10^{\frac{A_{S_{dB}}}{20}} = 1.5$$

Fase del guadagno asintotico per $\omega=\omega_S$:

$$\Delta\varphi_S = 90° - 45°\cdot\log\left(\frac{\omega_S}{0.1\,\omega_{P_1}}\right) = 31.5\cdot°$$

$$V_O = A_S\cdot V_S = 7.5\,V \qquad \varphi_O = \varphi_S + \Delta\varphi_S = 76.5\cdot°$$

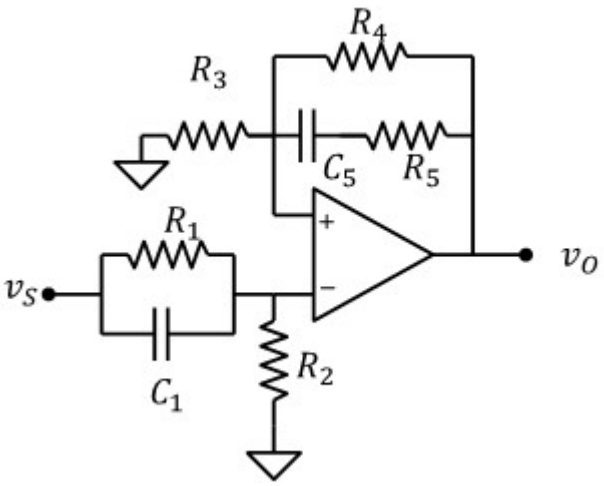
$$v_O(t) = V_O\cdot\sin(\omega_S\cdot t + \varphi_O)$$



Esercizio 15

DATI:

$R_2 = 1k\Omega, R_3 = 1k\Omega, R_4 = 99k\Omega\ C_1 = 11\,1nF,\ C_5 = 918nF$



1) calcolare R_1 e R_2 affinché $R_{IN} = 10k\Omega$ in ω=0

e |W(0)| = |W(∞)|

$R_{IN} = R_1 + R_2$ $\quad$ $\boxed{R_1 = R_{IN} - R_2 = 9\cdot k\Omega}$

Guadagno in ω=0 $\quad A_0 = \dfrac{R_2}{R_1 + R_2}\cdot\left(1 + \dfrac{R_4}{R_3}\right) = 10$

Guadagno in ω=∞ $\quad A_{HF} = \left(1 + \dfrac{R_P}{R_3}\right)$

con: $\quad R_P = \dfrac{R_4\cdot R_5}{R_4 + R_5}$ $\quad R_P = (A_0 - 1)\cdot R_3 = 9\cdot k\Omega$ $\quad \boxed{R_5 = \left(\dfrac{1}{R_P} - \dfrac{1}{R_4}\right)^{-1} = 9.9\cdot k\Omega}$

2) Funzione di trasferimento

Calcoliamo le impedenze: $\quad Z_1 = \dfrac{R_1}{1 + j\omega\cdot R_1\cdot C_1}$ $\quad Z_5 = \dfrac{1 + j\omega\cdot R_5\cdot C_5}{j\omega\cdot C_5}$ $\quad Z_4 = \dfrac{R_4\cdot Z_5}{R_4 + Z_5} = \dfrac{R_4\cdot(1 + j\omega\cdot R_5\cdot C_5)}{1 + j\omega\cdot(R_5 + R_4)\cdot C_5}$

Potenziale del terminale non invertente: $\quad v_P = v_S\cdot\dfrac{R_2}{R_2 + Z_1} = v_S\cdot\dfrac{R_2}{R_1 + R_2}\cdot\dfrac{(1 + j\omega\cdot R_1\cdot C_1)}{1 + j\omega\cdot\dfrac{R_2\cdot R_1}{R_1 + R_2}\cdot C_1}$

Nodo di uscita:

$$v_O = v_P\cdot\left(1 + \frac{Z_4}{R_3}\right) = v_P\cdot\left[1 + \frac{1}{R_3}\cdot\frac{R_4\cdot(1 + j\omega\cdot R_5\cdot C_5)}{1 + j\omega\cdot(R_5 + R_4)\cdot C_5}\right] = v_P\cdot\left[\frac{1}{R_3}\cdot\frac{R_3 + R_4 + j\omega\cdot(R_5\cdot R_3 + R_4\cdot R_3 + R_5\cdot R_4)\cdot C_5}{1 + j\omega\cdot(R_5 + R_4)\cdot C_5}\right]$$

$$v_O = v_P\cdot\left[\frac{R_3 + R_4}{R_3}\cdot\frac{1 + j\omega\cdot\left(R_5 + \dfrac{R_3\cdot R_4}{R_3 + R_4}\right)\cdot C_5}{1 + j\omega\cdot(R_5 + R_4)\cdot C_5}\right]$$

$$v_O = \frac{R_2}{R_1 + R_2}\cdot\frac{(1 + j\omega\cdot R_1\cdot C_1)}{1 + j\omega\cdot\dfrac{R_2\cdot R_1}{R_1 + R_2}\cdot C_1}\cdot\frac{1 + j\omega\cdot\left(R_5 + \dfrac{R_4\cdot R_3}{R_3 + R_4}\right)\cdot C_5}{1 + j\omega\cdot(R_5 + R_4)\cdot C_5}\cdot v_S$$

$$\frac{R_2\cdot R_1}{R_1 + R_2} = 900\,\Omega \qquad R_4 + R_5 = 108.9\cdot k\Omega \qquad R_5 + \frac{R_4\cdot R_3}{R_3 + R_4} = 10.89\cdot k\Omega$$

Poniamo: $A = \frac{R_2}{R_1 + R_2} \cdot \frac{R_3 + R_4}{R_3} = 10$ che coincide con il guadagno per ω=0

$$\omega_{Z_1} = \left[\left(R_5 + \frac{R_4 \cdot R_3}{R_3 + R_4}\right) \cdot C_5\right]^{-1} = 100 \cdot s^{-1}$$

$$\omega_{Z_2} = \left(R_1 \cdot C_1\right)^{-1} = 1 \times 10^3 \cdot s^{-1}$$

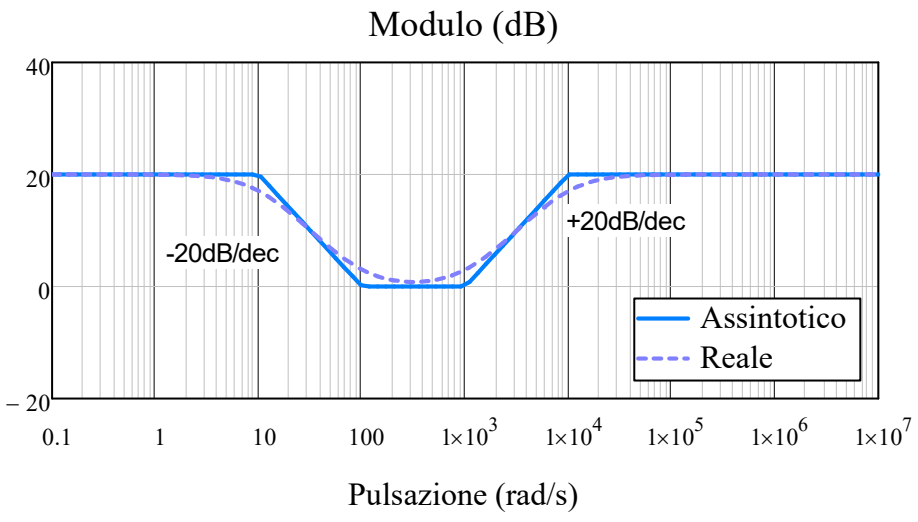
$$\omega_{P_1} = \left[\left(R_5 + R_4\right) \cdot C_5\right]^{-1} = 10 \cdot s^{-1}$$

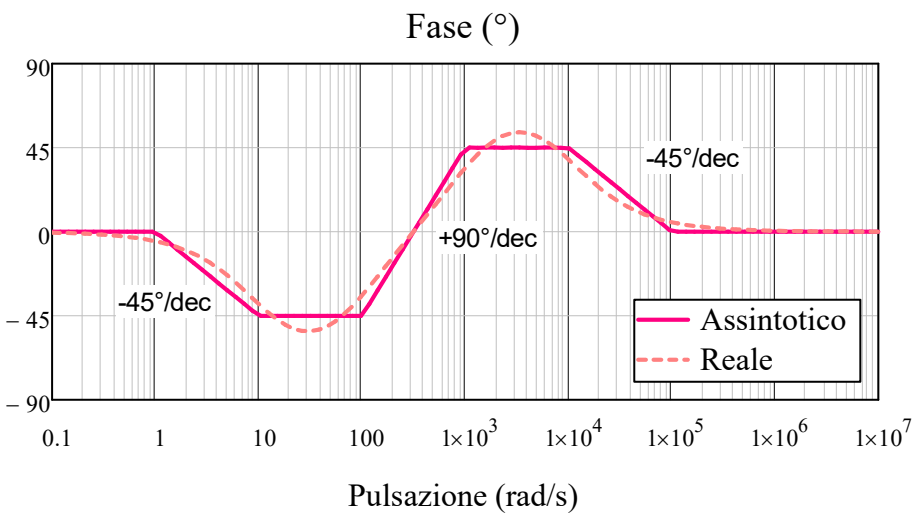
$$\omega_{P_2} = \left(\frac{R_2 \cdot R_1}{R_1 + R_2} \cdot C_1\right)^{-1} = 1 \times 10^4 \cdot s^{-1}$$

Otteniamo la seguente funzione di trasferimento:

$$W(\omega) = A \cdot \frac{\left(1 + \frac{j\omega}{\omega_{Z_1}}\right) \cdot \left(1 + \frac{j\omega}{\omega_{Z_2}}\right)}{\left(1 + \frac{j\omega}{\omega_{P_1}}\right) \cdot \left(1 + \frac{j\omega}{\omega_{P_2}}\right)}$$

3) Diagramma di Bode asintotico del modulo e della fase





5) Ampiezza e fase del segnale di uscita, con segnale di ingresso:

$V_{S1} = 10mV, \quad \omega_1 = 10^6 s^{-1}, \quad \varphi_1 = 180°$

$V_{S2} = 1mV, \quad \omega_2 = 690 s^{-1}, \quad \varphi_2 = 90°$

$v_S(t) = V_{S1} \cdot \sin(\omega_1 \cdot t + \varphi_1) + V_{S2} \cdot \sin(\omega_2 \cdot t + \varphi_2)$

Guadagno in ω_1

$W_1 = 20dB \quad A_1 = 10^{\frac{W_1}{20}} = 10$

$\Delta\varphi_1 = 0°$

$V_{O1} = A_1 \cdot V_{S1} = 0.1\,V$

$\varphi_{O1} = \varphi_1 + \Delta\varphi_1 = 180 \cdot °$

Guadagno in ω_2

$W_2 = 0dB \quad A_2 = 10^{\frac{W_2}{20}} = 1$

$\Delta\varphi_2 = -45° + 90° \cdot \log\left(\frac{\omega_2}{\omega_{Z_1}}\right) = 30 \cdot °$

$V_{O2} = A_2 \cdot V_{S2} = 1 \cdot mV$

$\varphi_{O2} = \varphi_2 + \Delta\varphi_2 = 120 \cdot °$

$v_O(t) = V_{O1} \cdot \sin(\omega_1 \cdot t + \varphi_{O1}) + V_{O2} \cdot \sin(\omega_2 \cdot t + \varphi_{O2})$

Esercizio 16

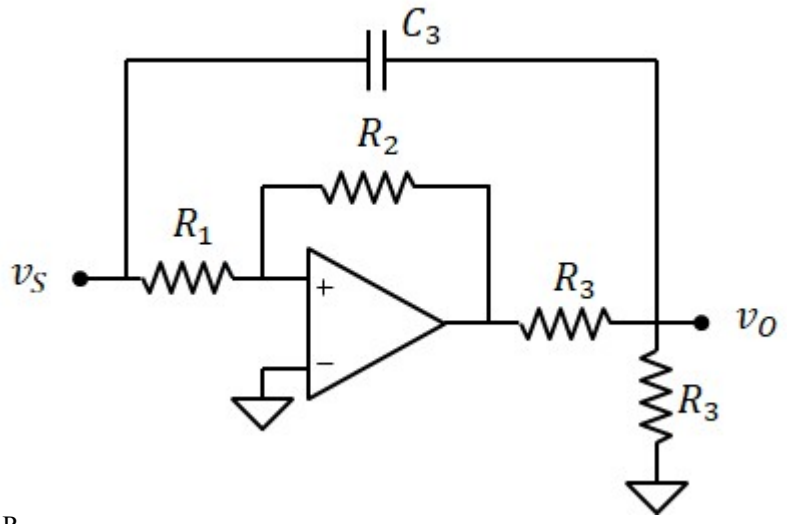
DATI:

$R_1 = 1k\Omega$

$R_2 = 20k\Omega$

$R_3 = 5k\Omega$

$C_3 = 400nF$



1) Funzione di trasferimento

Tensione di uscita dell'operazionale: $v_A = \frac{-R_2}{R_1} \cdot v_S$

Tensione di uscita (legge di kirchhoff):

$$\frac{v_O}{R_3} + (v_O - v_S) \cdot j\omega \cdot C_3 + \frac{v_O - v_A}{R_3} = 0$$

$$\frac{v_O}{R_3} + j\omega \cdot C_3 \cdot v_O + \frac{v_O}{R_3} = \frac{v_A}{R_3} + j\omega \cdot C_3 \cdot v_S$$

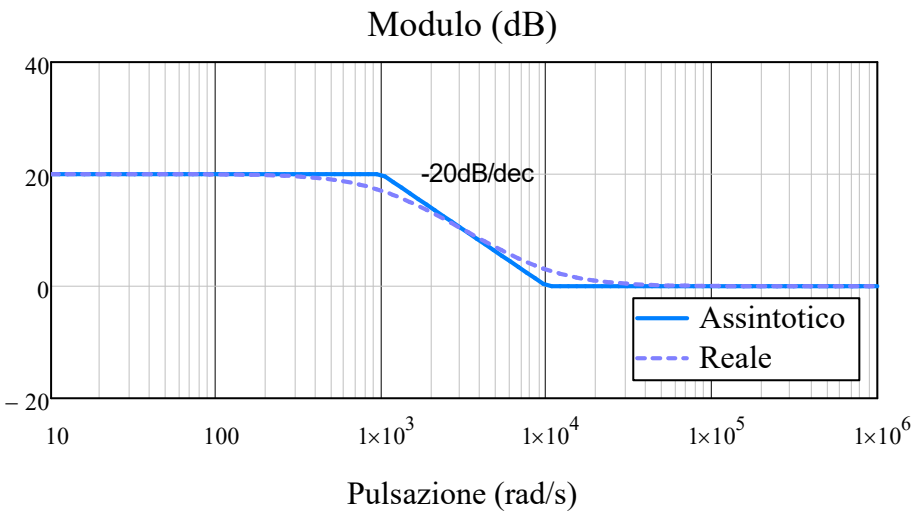
$$v_O = \frac{R_3}{2 + j\omega \cdot C_3 \cdot R_3} \cdot \left(\frac{v_A}{R_3} + j\omega \cdot C_3 \cdot v_S \right) = \frac{R_3}{2 + j\omega \cdot C_3 \cdot R_3} \cdot \left(\frac{-R_2}{R_1 \cdot R_3} + j\omega \cdot C_3 \right) \cdot v_S = \frac{R_3}{2 + j\omega \cdot C_3 \cdot R_3} \cdot \left(\frac{-R_2 + j\omega \cdot C_3 \cdot R_1 \cdot R_3}{R_1 \cdot R_3} \right) \cdot v_S$$

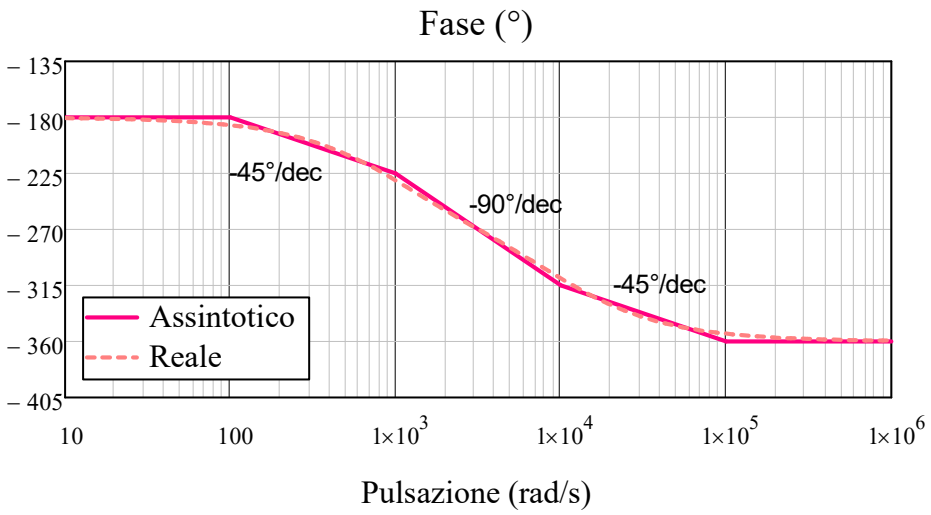
$$\frac{v_O}{v_S} = \frac{-R_2}{2 \cdot R_1} \cdot \frac{\left(1 - j\omega \cdot C_3 \cdot R_1 \cdot \frac{R_3}{R_2} \right)}{1 + j\omega \cdot \frac{C_3 \cdot R_3}{2}} \cdot v_S$$

$\omega_{P_1} = \frac{2}{R_3 \cdot C_3} = 1 \times 10^3 \, s^{-1}$ $\quad \omega_{Z_1} = \frac{R_2}{C_3 \cdot R_1 \cdot R_3} = 1 \times 10^4 \, s^{-1}$ (zero a parte reale positiva) $\quad A = \frac{-R_2}{2 \cdot R_1} = -10$

$$\boxed{W(\omega) = A \cdot \frac{\left(1 - \frac{j\omega}{\omega_{Z_1}} \right)}{\left(1 + \frac{j\omega}{\omega_{P_1}} \right)}}$$

2) diagramma di Bode asintotico del modulo e della fase





3) Ampiezza e fase del segnale di uscita, con segnale di ingresso:

$V_{S1} = 1V, \qquad \omega_1 = 2\cdot 10^3 s^{-1}, \quad \varphi_1 = 45°$

$v_S(t) = V_{S1}\cdot \sin(\omega_1 \cdot t + \varphi_1)$

Guadagno in ω_1 $\qquad W_1 = 20dB - 20dB\cdot \log\left(\frac{\omega_1}{1000s^{-1}}\right) = 13.979 \qquad A_1 = 10^{\frac{W_1}{20}} = 5$

Fase in ω_1: $\qquad \Delta\varphi_1 = -180° - 45° - 90°\cdot \log\left(\frac{\omega_1}{1000s^{-1}}\right) = -252\cdot °$

$\boxed{V_{O1} = A_1\cdot V_{S1} = 5\,V} \qquad \boxed{\varphi_{O1} = \varphi_1 + \Delta\varphi_1 = -207\cdot °}$

$\boxed{v_O(t) = V_{O1}\cdot \sin(\omega_1 \cdot t + \varphi_{O1})}$

Esercizio 17

DATI:

$R_1 = 10k\Omega$,

$R_2 = 200k\Omega$,

$R_3 = 1k\Omega$,

$R_4 = 5k\Omega$

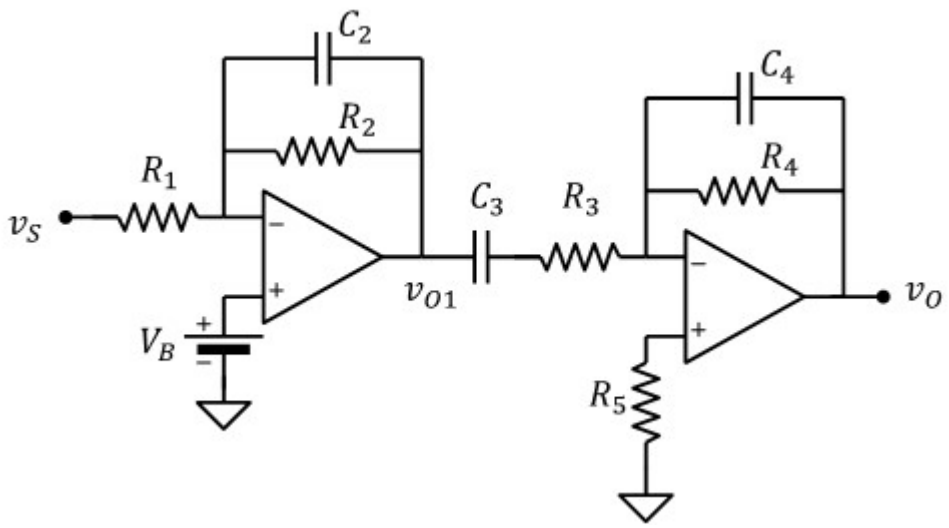
$R_5 = 3.3k\Omega$

$C_2 = 50pF$,

$C_3 = 10\mu F$,

$C_4 = 20nF$,

$V_B = 1V$



1) resistenza di ingresso

$R_{IN} = R_1 = 10 \cdot k\Omega$

2) La tensione di uscita in condizioni stazionarie con $v_S = 200mV$

Tutte le capacità sono circuiti aperti.

$v_O = 0$

3) Funzione di trasferimento

Applichiamo la sovrapposizione degli effetti:
con $v_S = 0V$ e V_B ON

$$v_{O1} = V_B \cdot \left(1 + \frac{Z_2}{R_2}\right)$$

poichè VB è costante, l'uscita è costante pari a:

$$v_{O1} = V_B \cdot \left(1 + \frac{R_2}{R_1}\right) = 21\,V$$

il secondo stadio in presenza di un ingresso costante ha $v_O = 0$ poichè all suo ingresso la capacità C_3 è in aperto.
Di conseguenza $v_N = v_P = v_O = 0$

con $V_B = 0$ e v_S ON:

Primo stadio

$$Z_2 = \frac{R_2}{1 + j\omega \cdot R_2 \cdot C_2}$$

$$v_{O1} = -v_S \cdot \frac{Z_2}{R_1} = \frac{-R_2}{R_1} \cdot \frac{1}{1 + j\omega \cdot R_2 \cdot C_2} \cdot v_S$$

Secondo stadio

$$v_O = -\frac{Z_4}{Z_3} \cdot v_{O1} \qquad Z_3 = \frac{1 + j\omega \cdot R_3 C_3}{j\omega \cdot C_3} \qquad Z_4 = \frac{R_4}{1 + j\omega \cdot R_4 \cdot C_4}$$

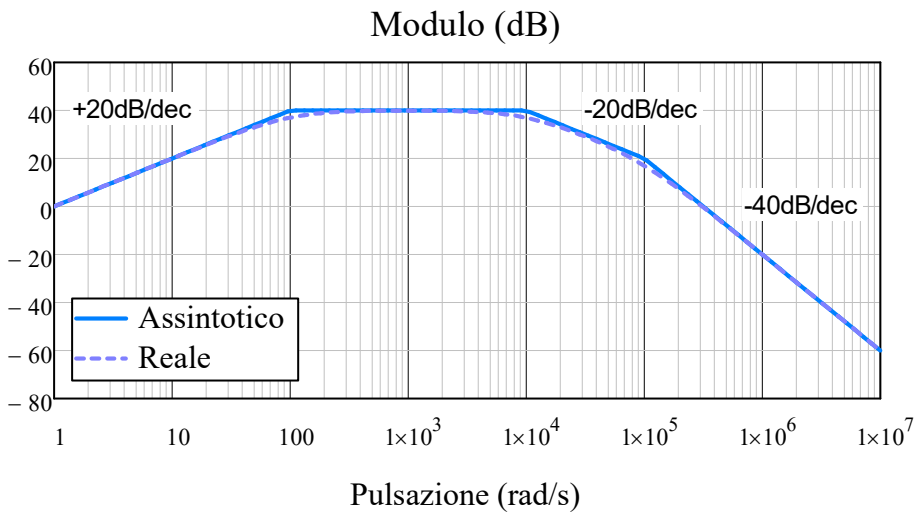
$$v_O = -\frac{j\omega \cdot C_3 \cdot R_4}{\left(1 + j\omega \cdot R_3 C_3\right)\cdot\left(1 + j\omega \cdot R_4 \cdot C_4\right)} \cdot v_{O1} = \frac{j\omega \cdot C_3 \cdot R_4}{\left(1 + j\omega \cdot R_3 C_3\right)\cdot\left(1 + j\omega \cdot R_4 \cdot C_4\right)} \cdot \left(\frac{R_2}{R_1} \cdot \frac{1}{1 + j\omega \cdot R_2 \cdot C_2} \cdot v_S\right)$$

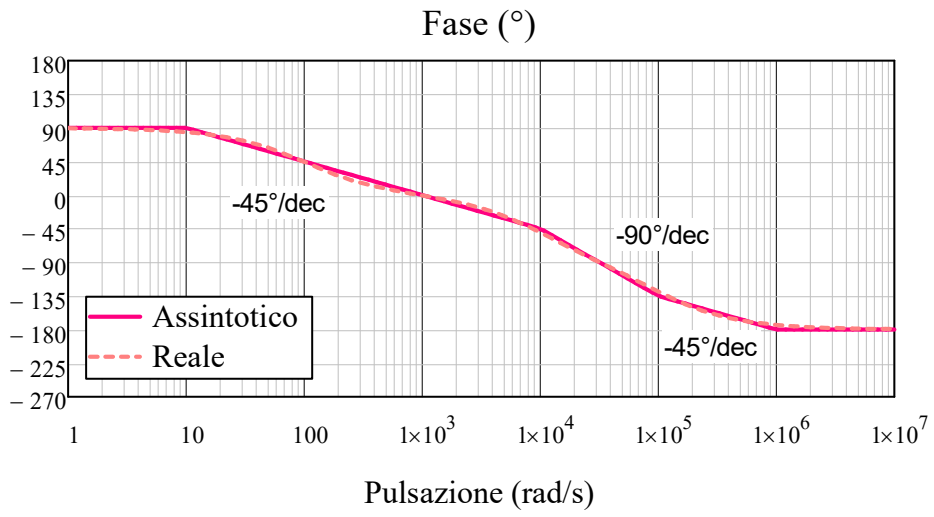
$$v_O = \frac{R_2 \cdot R_4}{R_1 \cdot R_3} \cdot \frac{j\omega \cdot R_3 \cdot C_3}{\left(1 + j\omega \cdot R_2 \cdot C_2\right)\cdot\left(1 + j\omega \cdot R_3 C_3\right)\cdot\left(1 + j\omega \cdot R_4 \cdot C_4\right)} \cdot v_S$$

Poniamo: $A = \frac{R_2 \cdot R_4}{R_1 \cdot R_3} = 100$ $\quad \omega_{P_1} = \frac{1}{R_3 \cdot C_3} = 100 \cdot s^{-1}$ $\quad \omega_{P_2} = \frac{1}{R_4 \cdot C_4} = 1 \times 10^4 \cdot s^{-1}$ $\quad \omega_{P_3} = \frac{1}{R_2 \cdot C_2} = 1 \times 10^5 \cdot s^{-1}$

$$W(\omega) = A \cdot \frac{j\frac{\omega}{\omega_{P_1}}}{\left(1 + j\frac{\omega}{\omega_{P_1}}\right)\cdot\left(1 + j\frac{\omega}{\omega_{P_2}}\right)\cdot\left(1 + j\frac{\omega}{\omega_{P_3}}\right)}$$

4) Diagramma di Bode asintotico del modulo e della fase





5) Ampiezza e fase del segnale di uscita, con segnale di ingresso:

$V_{S0} = 1V$

$V_{S1} = 10mV, \quad \omega_1 = 10^3 s^{-1}, \quad \varphi_1 = 180°$

$V_{S2} = 300mV, \quad \omega_2 = 10^6 s^{-1}, \quad \varphi_2 = 90°$

$v_S(t) = V_{S0} + V_{S1} \cdot \sin(\omega_1 \cdot t + \varphi_1) + V_{S2} \cdot \sin(\omega_2 \cdot t + \varphi_2)$

Guadagno in DC (ω=0): $W_0 = 0$ $\quad V_{O0} = W_0 \cdot V_{S0} = 0$

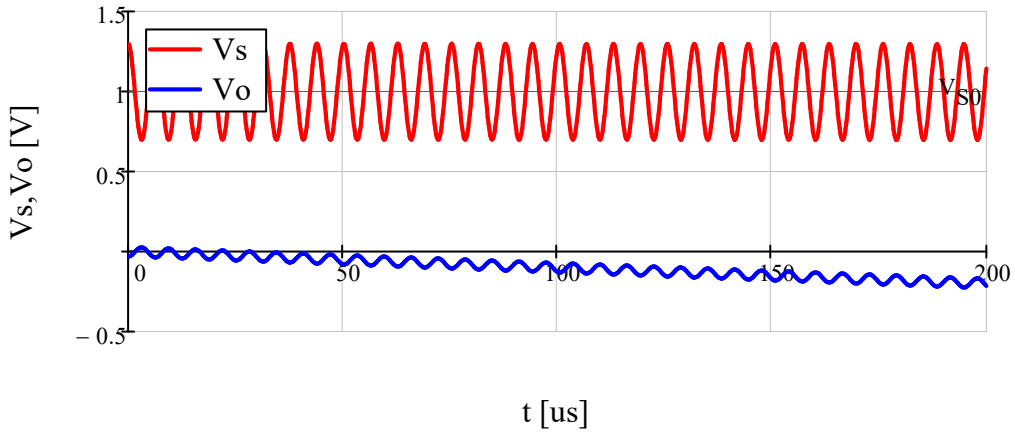
Guadagno in ω_1: $W_1 = 40dB \quad A_1 = 10^{\frac{W_1}{20}} = 100$ $\quad V_{O1} = A_1 \cdot V_{S1} = 1\,V$

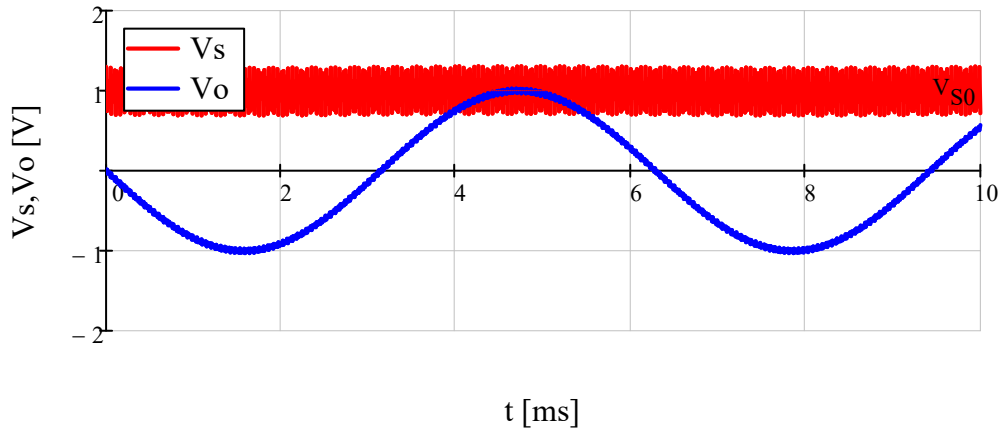
$\Delta\varphi_1 = 0°$ $\quad \varphi_{O1} = \varphi_1 + \Delta\varphi_1 = 180 \cdot °$

Guadagno in ω_2: $W_2 = -20dB \quad A_2 = 10^{\frac{W_2}{20}} = 0.1$ $\quad V_{O2} = A_2 \cdot V_{S2} = 30 \cdot mV$

$\Delta\varphi_2 = -180°$ $\quad \varphi_{O2} = \varphi_2 + \Delta\varphi_2 = -90 \cdot °$

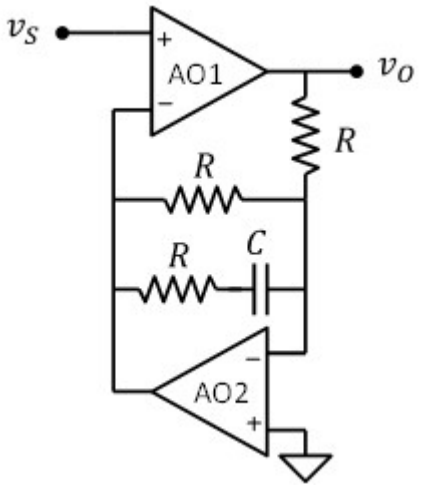
$v_O(t) = V_{O0} + V_{O1} \cdot \sin(\omega_1 \cdot t + \varphi_{O1}) + V_{O2} \cdot \sin(\omega_2 \cdot t + \varphi_{O2})$





Esercizio 18

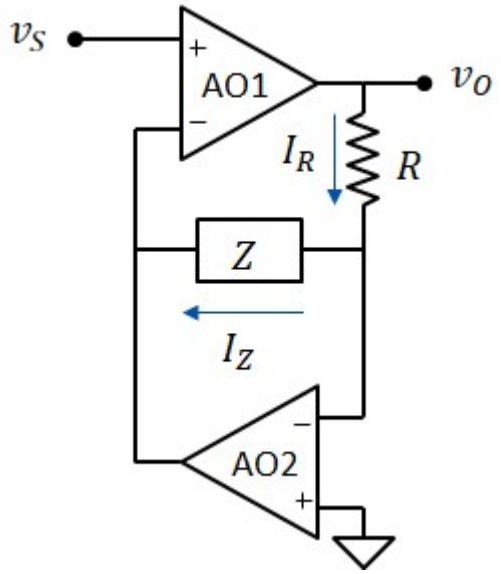
DATI: $R = 10k\Omega$, $C = 100nF$



1) Funzione di trasferimento

Impedenza della rete di retroazione comune ai due operazionali

$$Z = \frac{R\left(R + \frac{1}{i\omega \cdot C}\right)}{R + R + \frac{1}{j\omega \cdot C}} = R \cdot \frac{1 + i\omega \cdot R \cdot C}{1 + i\omega \cdot 2R \cdot C}$$



Legge di kirchhoff tra R e Z:

$$I_R = I_Z$$

$$\frac{v_{O1} - v_{N2}}{R} = \frac{v_{N2} - v_{N1}}{Z}$$

Per il principio del cortocircuito virtuale:

$$v_{N1} = v_S \qquad v_{N2} = 0$$

$$\frac{v_{O1}}{R} = \frac{-v_S}{Z}$$

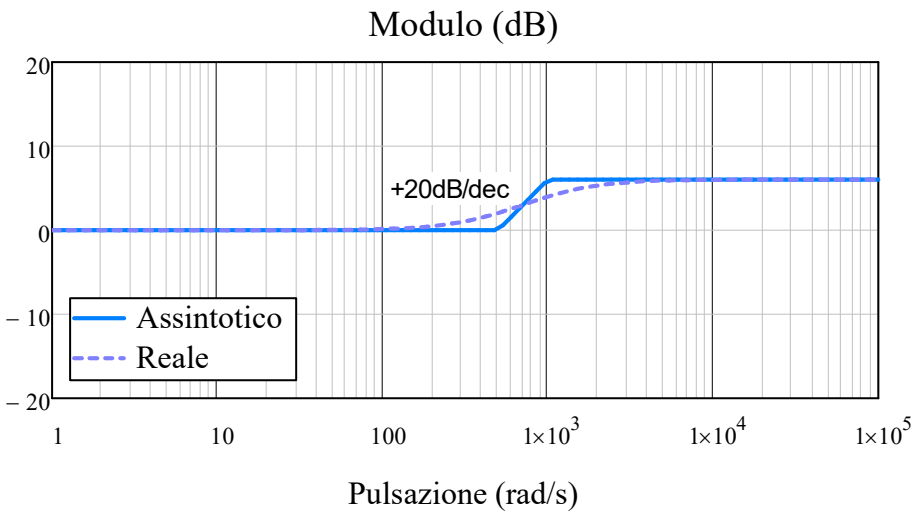
$$v_O = v_{O1} = \frac{-R}{Z} \cdot v_S$$

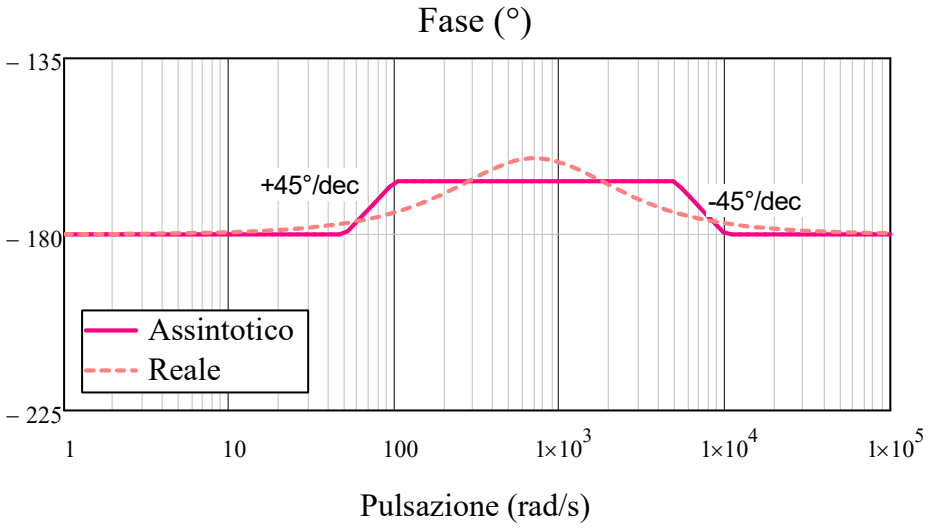
$$W(\omega) = \frac{-R}{Z} = \frac{1 + i\omega \cdot 2R \cdot C}{1 + i\omega \cdot R \cdot C}$$

$$\boxed{\omega_{Z_1} = \frac{1}{2 \cdot R \cdot C} = 500 \cdot s^{-1}} \qquad \boxed{\omega_{P_1} = \frac{1}{R \cdot C} = 1 \times 10^3 \cdot s^{-1}}$$

$$\boxed{W(\omega) = -\frac{1 + \frac{j\omega}{\omega_{Z_1}}}{1 + \frac{j\omega}{\omega_{P_1}}}}$$

2) Tracciare il diagramma di Bode asintotico del modulo e della fase

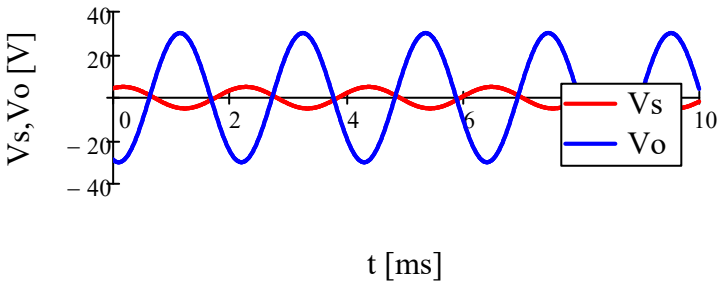




3) Ampiezza e fase del segnale di uscita, quando all'ingresso è presente il segnale: $v_S = V_S \cdot \sin(\omega_1 \cdot t + \varphi_1)$

con $V_S = 5V$, $\omega_1 = 3 \cdot 10^3 s^{-1}$, $\varphi_1 = 60°$

Modulo del guadagno asintotico per $\omega=\omega_1$: $W_1 = 0 + 20 \cdot \log\left(\frac{\omega_1}{\omega_{Z_1}}\right) = 15.6 \cdot dB$ $A_1 = 10^{\frac{W_1}{20}} = 6$ $\boxed{V_O = A_1 \cdot V_S = 30\,V}$

Fase del guadagno asintotico per $\omega=\omega_1$: $\Delta\varphi_1 = -180° + 45° \cdot \left(\log\left(\frac{\omega_{P_1} \cdot 0.1}{\omega_{Z_1} \cdot 0.1}\right)\right) = -166 \cdot °$ $\boxed{\varphi_O = \varphi_1 + \Delta\varphi_1 = -106 \cdot °}$

$v_S(t) = V_S \cdot \sin(\omega_1 \cdot t + \varphi_1)$ $v_O(t) = V_O \cdot \sin(\omega_1 \cdot t + \varphi_O)$